Why Cluster Validity Indices Mislead: Misranking Analysis and the Arboris Distance

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Abstract

Cluster validity indices (CVIs) are widely used to assess clustering quality, compare clustering algorithms, and estimate the number of clusters. However, many CVIs rely on Euclidean notions of compactness and separation, which can favour geometrically simple partitions over the structure intended by the data. In this work, we examine whether 29 internal CVIs assign their optimal scores to meaningful reference labelsets across 44 synthetic benchmark datasets and 8 real-world datasets. Using both handcrafted labelsets and labelsets produced by clustering algorithms, we show that many CVIs frequently misrank the ground-truth labels and, in some cases, the algorithm-generated labelsets with the highest Adjusted Rand Index (ARI). These failures are particularly pronounced for irregular cluster shapes, imbalanced clusters, and overlapping clusters. To address this geometry mismatch, we propose *Arboris Distance* (**AD**), i.e., the distance of the tree, a minimum-spanning-tree-based distance that captures data connectivity rather than relying only on Euclidean distances. We use **AD** to extend three widely used CVIs, namely Silhouette, Davies-Bouldin, and Calinski-Harabasz, and introduce the *Arboris index*, a new internal CVI. Across the synthetic benchmarks, AD-based CVIs reduce misrankings and identify the reference clustering more often than the evaluated baselines.

Keywords: cluster validity index (CVI); internal cluster validity index; external cluster validity index; Arboris Distance; graph-based distance; minimum spanning tree; non-globular clusters; imbalanced clustering

1 Introduction

Clustering is an unsupervised machine learning task that groups data points according to similarity, with applications in domains such as data mining, pattern recognition, bioinformatics, and market analysis [1, 2]. Since class labels are unavailable, clustering typically aims to maximize intra-cluster similarity and minimize inter-cluster similarity [3, 4], making the evaluation of the resulting partition inherently difficult [5, 6]. This difficulty is compounded by the combinatorial nature of clustering, as many clustering formulations are NP-hard and the number of possible assignments for N data points and K clusters grows on the order of K^N [7–10]. In practice, clustering outcomes are also affected by dataset size, overlap, imbalance, dimensionality, noise, and the assumed number of clusters [11–13].

The most common approach for evaluating the performance of a clustering algorithm is to use Cluster Validity Indices (CVIs) [3, 5, 11, 12, 14–19].

CVIs quantify properties such as compactness/cohesion, often measured through *intra-cluster* distances [19–21], and separation/heterogeneity, often measured through *inter-cluster* distances, to estimate how well a partition reflects the data structure [5, 15–18, 21].

Many CVIs were designed to address specific data characteristics, especially compact, well-separated globular clusters, and are therefore data-dependent [12, 13, 21].

However, not all datasets contain convex structures, and such CVIs may fail to correctly assess clustering performance when irregular shapes are present [22–24]. There are many domains in which non-convex shapes are common, including geospatial data [25], animal movements [26], cosmology [27], spatial omics [28], social and biological networks [29]. Consequently, CVI choice can directly affect algorithm comparison, parameter selection, and the interpretation of clustering results.

CVIs can be categorized as *external* or *internal* [3, 18, 24, 30]. External indices compare clustering labels with reference labels, which limits their use to datasets with available ground truth [31–33]. Additionally, they may be affected by biases related to the number of clusters or class imbalance [34–36]. In contrast, internal indices evaluate a clustering result using only the data and the assigned labels, usually through properties such as compactness, separation, structure, or shape [20, 32, 33, 37].

Because many internal CVIs measure compactness and separation through Euclidean distances, they may implicitly favour dense, well-separated, globular partitions [12, 22, 24].

As a result, an incorrect but geometrically simpler partition can receive a better score than a structurally meaningful one, especially for non-convex clusters.

Several CVIs have been proposed for irregular shapes, including density- and graph-based indices [22, 38–44], but their applicability remains limited across datasets with different shapes, densities, overlap levels, and cluster sizes. This limitation is practically important because internal CVIs are often used not only to report clustering quality, but also to drive decisions such as estimating the number of clusters and selecting algorithm parameters. For example, in K-Means, the value of K is commonly varied and the solution that optimizes a chosen CVI score is selected [15, 20, 21, 24, 45, 46]. If the CVI itself favours a geometrically convenient but structurally incorrect partition, this procedure can select the wrong number of clusters or

favour an inferior parametrization. Consistently with this concern, CVI scores can depend strongly on the number of clusters; in one study, only 14 of 68 internal CVIs were invariant to this factor [20].

Another common use case of CVIs is the comparison of newly developed clustering algorithms with existing methods [15, 21, 47]. However, different CVIs optimize different, and sometimes conflicting, notions of clustering quality [18]. As a result, they may rank the same candidate partitions differently, and no single CVI is uniformly reliable across datasets and clustering scenarios [12, 31, 45, 48–50]. This creates a practical risk: if a CVI assigns a more favourable score to an incorrect partition than to a meaningful reference partition, both algorithm comparison and parameter selection can become biased.

To examine this failure mode directly, we extend the usual benchmark-based analysis with a labelset-based evaluation. For each dataset, we compare the ground-truth labels with several alternative labelsets, including random labels and simple linear separations of the feature space. These handcrafted labelsets are designed to be structurally simple but generally incorrect, so a reliable CVI should rank the ground-truth labels above them. We also evaluate labelsets produced by clustering algorithms, allowing us to test whether CVIs identify the best algorithm-generated partition rather than only the ground truth.

Motivated by the geometry mismatch behind many of these failures, we propose Arboris Distance (AD), a graph-based distance computed from paths in a minimum spanning tree built on a nearest-neighbor graph. AD captures the connectivity structure of the data and reduces the Euclidean shortcut effect that can distort intra-cluster distances in non-convex clusters [37]. We use AD to extend Euclidean-distance-based CVIs and introduce a new AD-based index. The specific novelty lies in using the AD metric to systematically resolve surrogate misrankings in internal CVIs without changing the internal index aggregation logic itself.

In this work, we evaluate the ranking performance of 29 internal CVIs and compare them using both handcrafted labelsets and labelsets obtained from commonly used clustering algorithms on 44 synthetic benchmark datasets with diverse characteristics. The goal is to determine when CVIs assign more favourable scores to incorrect or suboptimal labelsets, and to assess whether geometry-aware distance computation can reduce these failures.

The main contributions of this work are:

- Systematic comparative analysis: an evaluation of 29 internal CVIs (traditional and recently developed) across synthetic datasets with varying shape, overlap, imbalance, noise, and dimensionality.
- Analysis of CVI failure modes: an empirical study of the conditions under which CVIs assign more favourable scores to incorrect or suboptimal labelsets. Multiple scenarios are considered:
  - Labelset-based evaluation protocol: Handcrafted labelsets that expose simple misranking cases, a methodology for testing whether CVIs rank meaningful reference labelsets above simple, structurally incorrect alternatives.;
  - Overlap and imbalance analyses to identify breaking points in CVI behaviour;

  - Number-of-clusters estimation through CVI-guided parameter selection;
  - Aggregated analyses across 44 synthetic clustering benchmark datasets;
  - Statistical validation of the obtained results.
  - Real-world correlation analysis: an additional analysis on 8 real-world datasets using correlations between internal CVI scores and Adjusted Rand Index values across clustering algorithms and parameterizations.

- Arboris Distance (AD): a novel minimum-spanning-tree-based distance computed on nearest-neighbor graphs to capture data connectivity.
- AD-based CVI extensions: extensions of three widely used internal CVIs, namely Silhouette, Davies-Bouldin, and Calinski-Harabasz.
- Arboris index: a new internal CVI based on Arboris Distance.
- *(Supplementary, see Section B)* Time analysis: comparative analysis by number of samples and number of dimensions across all CVIs.
- *(Supplementary, see Section C)* AD-based clustering: K-Means extension using the Arboris distance allowing for clusterization of irregular shapes.
- *(Supplementary, see Section D)* Balanced external CVIs: variants of external CVIs designed to improve evaluation on imbalanced clustering.

The remainder of this paper is organized as follows. Section 2 (Materials and Methods) introduces the notation and formally defines the 29 internal CVIs evaluated in this work, formalizes the misranking problem underlying internal CVIs, describes the Arboris Distance and its metric properties, and presents the datasets, labelsets, external CVIs, and imbalance/overlap measures used throughout the analyses. Section 3 (Results) presents the experimental findings: we first examine the effects of imbalance and overlap on CVI behavior, then analyze the use of internal CVIs for estimating the number of clusters, followed by a direct comparison of classical CVIs with their Edging Distance and Arboris Distance extensions. The analysis continues with an aggregated ranking analysis across all 44 synthetic benchmark datasets, and finally correlation analyses on real-world UCI and image datasets. Section 4 (Conclusions) summarizes the main findings, discusses the limitations of the proposed approach, and outlines directions for future work. Supplementary material includes a timing analysis of all CVIs, an application of Arboris Distance to K-Means clustering, balanced variants of external CVIs, and detailed mathematical formulations of all evaluated indices.

2 Materials and Methods

2.1 Cluster validity indices

We use the notations summarized in Table 1, following common CVI conventions [51]. Let $X = \{x_i\}_{i=1}^{N}$ be a dataset containing N data points divided into K clusters $C_1, C_2, \ldots, C_K$. Each cluster C_k contains n_k points and has the cluster centre, i.e. centroid, μ_k, while $d_E(x_i, x_j)$ denotes the Euclidean distance between points x_i and x_j. When needed, μ_l denotes the centroid of the closest cluster distinct from C_k.

Many internal CVIs have been proposed, and several comparative studies have attempted to rank them across domains and clustering scenarios [11, 12, 18, 19, 52–55]. Table 2 summarizes the 29 internal CVIs considered in this work, covering classical

Table 1: Notations used in the equations.

Term / Notation	Meaning
$X = \{x_i\}_{i=1}^{N}$	Dataset containing N data points, $x_i \in \mathbb{R}^D$
N	The number of points in the dataset
D	The number of dimensions/features in the dataset
K	The number of clusters in the dataset
C_k, where $k \in [1, K]$	The k-th cluster
n_k, where $k \in [1, K]$	The number of points in cluster k
$\mu_k = \frac{1}{n_k} \sum_{x_k \in C_k} x_k$, where $k \in [1, K]$	Centroid of cluster k
$\mu_{\text{data}} = \frac{1}{N} \sum_{i=1}^{N} x_i$	Mean of the whole dataset X
$d_E(x_i, x_j)$	Euclidean distance between x_i and x_j
$d_{\text{AD}}(x_i, x_j)$	Arboris Distance between x_i and x_j
$d(\cdot, \cdot)$	Generic distance used by a CVI, when the metric is not specified
N_{noise}	The number of noise points in the dataset

compactness/separation measures, density- and shape-aware indices, and graph-based criteria. For each CVI, we report its time complexity, main evaluation principle, and score range. In particular, this selection includes widely used indices such as the Silhouette index (S), the Davies-Bouldin index (DB), and the Calinski-Harabasz index (CH), which are commonly used for clustering evaluation and parameter selection. It also includes density- and shape-aware indices, such as Density-Based Clustering Validation (DBCV), the CDbw index, and the Validity Index for Arbitrary-Shaped Clusters based on Kernel Density Estimation (VIASCKDE), which were proposed to better handle non-globular or density-based clustering results. Further details on the individual CVIs and their formulations are provided in the supplementary material, Section A.

Overall, previous studies report inconsistent CVI rankings across datasets and scenarios [11, 14, 18–20, 58], suggesting that aggregate benchmark performance alone is insufficient to understand when and why CVIs fail. Therefore, before introducing the proposed distance-based extensions, we first formalise the ranking problem underlying internal cluster validity indices.

2.1.1 Formal analysis of misranking in cluster validity indices

We next formalize the ranking problem underlying internal CVIs. The goal is to distinguish an ideal reference-based evaluator, which has access to ground-truth labels, from internal CVIs, which score candidate labelsets using only the data geometry and the assigned labels. This distinction clarifies when Euclidean-distance-based CVIs may misrank candidate labelsets and why replacing the underlying distance with Arboris Distance can mitigate this issue.

Using the notations in Table 1, let $y^\star \in \{1, \dots, K\}^N$ denote the ground-truth labelset, whenever available. For a candidate labelset $\hat{y} \in \{1, \dots, \hat{K}\}^N$, where $\hat{K}$ is the number of clusters in the candidate labelset, define the pairwise co-membership relations:

$$R^\star_{ij} = \mathbf{1}\{y^\star_i = y^\star_j\}, \quad \hat{R}_{ij} = \mathbf{1}\{\hat{y}_i = \hat{y}_j\}, \quad i < j, \tag{1}$$

Table 2: Descriptive table of internal CVIs [56, 57].

Name	Abbreviation	Complexity	Description
Davies-Bouldin	DB (↓)	$O(ND + K^2D)$	Average similarity ratio of each cluster to most similar cl
Calinski-Harabasz	CH (↑)	$O(ND)$	Ratio of between-cluster to within-cluster dispersion
Silhouette index	S (↑)	$O(N^2)$	Mean intra-cluster vs lowest mean inter-cluster distance
Ball Hall index	BH (↓)	$O(ND)$	Mean within-cluster variance normalized by K
Xie-Beni index	XB (↓)	$O(ND)$	Fuzzy intra-cluster scatter / min inter-centroid dist squa
Xie-Beni* index	XB*(↓)	$O(ND)$	Variant of Xie-Beni with modified separation term
Dunn index	$\mathcal{D}$ (↑)	$O(N^2)$	Min inter-cluster distance / max intra-cluster diameter
Sum of squared errors	SSE (↓)	$O(NKD)$	Total within-cluster sum of squares
Duda Hart index	DH (↓)	$O(N^2D)$	Ratio of avg pairwise within to avg pairwise between dist
Beale Index	B (↓)	$O(ND + N \log N)$	Ratio of within-cluster to between-cluster sum of squares
R-Squared index	RS (↑)	$O(NK)$	Fraction of total variance explained by clustering
DBCV	DBCV (↑)	$O(N^2D + N^2 \log N)$	Compares intra-cluster density to inter-cluster density
Hartigan Index	H (↓)	$O(ND)$	Change in explained variance when increasing K
Centroid-based S	cSIL (↑)	$O(N^2)$	Per-point silhouette using centroid distances
Generalized Dunn 43	GD43 (↑)	$O(N^2)$	Variant using specific inter-cluster/intra-cluster choices
Generalized Dunn 53	GD53 (↑)	$O(N^2)$	Variant with different distance/diameter definitions
Partition Separation	PS (↑)	$O(N^2D)$	Prototype separation relative to within-cluster spread
rCIP	rCIP (↓)	$O(N^2D^3)$	Cross-potential between representative distributions
Within–Between	WB (↓)	$O(N^2)$	Combined within-cluster scatter measure
Score Function	SF (↑)	$O(N^2)$	Composite of inter-centroid and intra-cluster scatter
SD validity index	SD (↓)	$O(ND)$	Combined scattering and separation term
S_Dbw index	SDbw (↓)	$O(ND + K^2D)$	Within-cluster scatter and density-based between term
C index	C (↓)	$O(N^2)$	Pairwise within distances vs theoretical extremes
I index	I (↑)	$O(N^2D)$	Compactness and separation relative to internal dispersio
CDbw index	CDbw (↑)	$O(NKD)$	Density-aware validity index with middle-point density
VIASCKDE	VIASCKDE (↑)	$O(N^2)$	KDE-weighted compactness and separation
Comp. Sep. Index	CS (↓)	$O(N^2)$	Ratio of aggregated intra to aggregated inter distances
COP index	COP (↓)	$O(N^2D)$	Compactness vs farthest neighboring cluster distance
Sym Index	Sym (↑)	$O(N^2D)$	Symmetry distance extension of the I index

where $\mathbf{1}\{\cdot\}$ is the indicator function. An ideal evaluator should assign the maximum score to the ground-truth labels and penalize deviations from them. One simple way to formalize this is through a weighted pairwise disagreement measure:

$$L^\star(\hat{y}; X, y^\star) = \sum_{i<j} \alpha_{ij}(X, y^\star)\, |R^\star_{ij} - \hat{R}_{ij}|, \tag{2}$$

where the weights satisfy

$$\alpha_{ij}(X, y^\star) > 0, \quad \sum_{i<j} \alpha_{ij}(X, y^\star) = 1. \tag{3}$$

The corresponding ideal score is then

$$J^\star(\hat{y}; X, y^\star) = 1 - L^\star(\hat{y}; X, y^\star). \tag{4}$$

By construction,

$$J^\star(y^\star; X, y^\star) = 1, \tag{5}$$

while any incorrect labelset obtains a smaller value. The score $J^\star$ is not an internal CVI; rather, it is a conceptual benchmark describing what a correct ranking of candidate labelsets should mean when the ground truth is known.

Internal CVIs do not have access to $y^\star$. Instead, they score a candidate labelset using only the data geometry and the assigned labels. A broad class of internal CVIs may be represented abstractly as:

$$V_d(\hat{y}; X) = \Phi\big(C^d_{\text{intra}}(X, \hat{y}), C^d_{\text{inter}}(X, \hat{y}), K(\hat{y}), N\big)\,, \tag{6}$$

where $d(x_i, x_j)$, $x_i, x_j \in X$ is the underlying distance, C^d_{intra} measures within-cluster compactness, C^d_{inter} measures between-cluster separation, and Φ combines these terms according to the specific index. This viewpoint covers a large family of internal CVIs, including indices such as Silhouette, Davies–Bouldin, Calinski–Harabasz, Dunn, and related variants, even though their exact aggregation mechanisms differ. The important aspect is that such indices inherit the geometry of the distance d used in their compactness and separation terms.

From this perspective, the relevant question is not whether an internal CVI reproduces the ground-truth labels explicitly, but whether it *ranks* candidate labelsets consistently with the ideal benchmark $J^\star$. In other words, if one candidate's labelset is closer to the ground truth than another, an appropriate internal CVI should also prefer it. This may be viewed as an order-consistency requirement between the ideal evaluator and the internal surrogate.

The main source of failure is a mismatch between the geometry induced by the distance used inside the CVI and the actual structure of the clusters. In particular, Euclidean-distance-based CVIs may fail when there exist many pairs of points belonging to the same ground-truth cluster that are far apart in Euclidean distance, or many pairs belonging to different ground-truth clusters that are close in Euclidean distance. This situation occurs naturally for non-globular, concentric, filamentary, bridge-connected, or manifold-like clusters. In such cases, the correct partition is not smooth with respect to Euclidean geometry.

This leads to a simple but important conclusion. If an internal CVI rewards small Euclidean within-cluster dispersion and large Euclidean between-cluster separation, then there may exist datasets and candidate labelsets for which a more globular but structurally incorrect partition receives a better score than the ground-truth labelset. The failure therefore stems not only from the algebraic form of the CVI, but also from the distance used to evaluate compactness and separation.

The role of Arboris Distance can be understood precisely within this framework. Rather than changing the compactness or separation logic itself, Arboris Distance changes the geometry in which these quantities are measured. Let $G = (\mathcal{V}, \mathcal{E})$ be a connected undirected weighted graph built from the dataset $X = \{x_i\}_{i=1}^N$, with vertex set $\mathcal{V} = \{1, \ldots, N\}$, where each vertex $i \in \mathcal{V}$ corresponds to the data point x_i. For each edge $(u, v) \in \mathcal{E}$, define the Euclidean edge weight:

$$w_{uv} = d_E(x_u, x_v). \tag{7}$$

For example, G may be taken as a symmetrized k-nearest neighbor graph. Let $\mathcal{T} = (\mathcal{V}, \mathcal{E}_{\mathcal{T}})$ be a minimum spanning tree of G. For any two vertices i and j in $\mathcal{T}$, let $\mathcal{P}_{ij}(\mathcal{T})$ denote the unique simple path in $\mathcal{T}$ connecting i and j. Then, for two points $x_i, x_j \in X$, the Arboris Distance is defined as:

$$d_{\mathrm{AD}}(x_i, x_j) = \max_{(u,v) \in \mathcal{P}_{ij}(\mathcal{T})} w_{uv}. \tag{8}$$

Thus, $d_{\mathrm{AD}}(x_i, x_j)$ is the largest Euclidean edge weight encountered along the unique path connecting x_i and x_j in $\mathcal{T}$.

If the true clusters are internally connected through paths with small bottlenecks, while paths between different clusters must cross larger bottlenecks, then the ground-truth labelset becomes well aligned with the geometry induced by d_{AD}. In this case, Euclidean shortcuts are reduced: points close only in ambient space but separated by a structural gap are no longer treated as near, while points on the same irregular cluster remain connected through low-bottleneck paths. Consequently, Arboris-based internal CVIs are expected to rank the ground-truth labelset, or labelsets closest to it, more favourably than Euclidean-based CVIs on datasets with non-globular structure.

Therefore, the proposed Arboris-based extensions should be interpreted not as entirely new validation principles, but as geometry-aware replacements of the Euclidean metric inside existing compactness and separation criteria. This explains why they can mitigate the misranking phenomenon observed for many classical CVIs on irregularly shaped datasets, without implying that any single CVI can be universally optimal across all scenarios.

2.1.2 Edging Distance and its cluster validity indices extensions

Edging Distance (ED) [37] is a graph-based distance computation method designed for irregular cluster shapes. It has been applied to clustering validation through extensions of three CVIs and to a K-Means variant [59]. Edging Distance computes the k-nearest neighbors, which has a time complexity of $O(ND + N \log N)$. The method performs several iterations, up to $O(N)$ in the worst case, invoking nearest neighbor searches on unvisited subsets of the dataset, which results in a total time complexity of $O(N^2 D + N^2 \log N)$ in the worst case, including the sorting of distances which contributes with an $O(N \log N)$ complexity. Additionally, the Edging Distance extension of metrics uses the dataset centre computation, which, due to the pairwise distance matrix, has a time complexity of $O(N^2 D)$.

The Silhouette extension of the Edging Distance (ED-S) computes cluster means, which in a scenario with balanced clusters would result in $O(N^2 DK)$, but in the worst case of having a single dominant cluster would lead to $O(N^2 D)$. ED-S makes $\sim N \times K$ calls to the edging distance, which results in a final time complexity of $O(KN^3 D + KN^3 \log N)$.

The Davies-Bouldin extension of the Edging Distance (ED-DB) uses the same cluster mean computation. It also computes the inter-cluster distances as the Edging Distances between all clusters, requiring $O(K^2)$ ED calls and yielding a time complexity of $O(K^2 N^2 D + K^2 N^2 \log N)$. Finally, the intra-cluster distances are computed as the distance from each point to its corresponding cluster centroid; thus,

the edge distance is computed for each sample, resulting in a time complexity of $O(N^3D + N^3 \log N)$. In the general case where $K \ll N$, the final time complexity is $O(N^3D + N^3 \log N)$.

The Calinski-Harabasz extension of the Edging Distance (ED-CH), as in the original CH index, computes all cluster means and the overall mean of the dataset, which has a time complexity of $O(N^2D)$. It computes the Edging Distance between all cluster means and the overall mean, resulting in a time complexity of $O(KN^2D+KN^2 \log N)$. The intra-cluster distances are computed as the edging distance between each sample and its cluster mean resulting in $O(N^3D+N^3 \log N)$. Thus, the total time complexity is $O(N^3D + N^3 \log N)$.

These high computational costs motivate the more efficient Arboris Distance extensions introduced next.

2.1.3 Arboris Distance

Following the formal analysis from Section 2.1.1, the failure of many internal CVIs can be interpreted as a geometry mismatch: compactness and separation are measured in Euclidean space even when the true partition is connectivity-based rather than globular.

We propose a new distance metric to mitigate the limitations of the Euclidean distance for irregular cluster shapes. The *Arboris Distance* (*AD*, from Latin *arboris*, meaning "of the tree") is a graph-based distance metric that can capture the topological structure of the data by computing distances using a Minimum Spanning Tree (MST). It can be considered the minimax path distance in the MST. Unlike Euclidean distance, which measures straight-line separation, d_{AD} follows the structure and connectivity of the data, making it suitable for irregular cluster shapes, clusters connected by thin bridges, and clusters with varying densities.

AD naturally adapts to varying densities. In dense regions, distances are smaller; in sparse regions, distances reflect the actual connectivity rather than arbitrary straight-line measurements. This also has implications for noise robustness. By constructing an MST through Prim's algorithm as shown in Algorithm 1, Arboris distance filters out noise and focuses on the core structure, as outliers affect only local edges rather than global distance computations. Moreover, as d_{AD} is the maximum edge of the path in the MST, it is sensitive to the largest gap along the path rather than the sum of small local, possibly noisy, variations.

For clustering validation, *AD* better captures intra-cluster cohesion by using the maximum edge along the MST path within a cluster, as shown in Algorithm 2, which reflects the worst-case internal separation. The inter-cluster separation is computed through MST paths between clusters, revealing the true connectivity, not just centroid distances.

$$G = (\mathcal{V}, \mathcal{E}), \quad \mathcal{V} = \{1, \ldots, N\}, \quad \mathcal{E} = \{e_{ij} = (i, j, w_{ij}) \mid w_{ij} = d_E(x_i, x_j),\ x_j \in \text{kNN}(x_i)\}. \tag{9}$$

Using Prim's algorithm, the MST is computed from the kNN graph:

$$\mathcal{E}_{\mathcal{T}} = \underset{\mathcal{E}' \subseteq \mathcal{E},\ |\mathcal{E}'|=N-1}{\arg\min} \sum_{(u,v)\in\mathcal{E}'} w_{uv}, \qquad \mathcal{T} = (\mathcal{V}, \mathcal{E}_{\mathcal{T}}). \tag{10}$$

The *AD* between data points i and j is:

$$d_{\mathrm{AD}}(x_i, x_j) = \max_{e\in\mathcal{P}_{ij}(\mathcal{T})} w_e. \tag{11}$$

Metric Properties

- **Identity**: *AD* computes distances based on the graph; the distance between a node and itself remains 0: $d_{\mathrm{AD}}(x_i, x_i) = 0$. Conversely, if the dataset contains no duplicate points, then $d_{\mathrm{AD}}(x_i, x_j) = 0$ implies $i = j$.
- **Symmetry**: *AD* computes distance using an MST, which means there exists a single path from i to j regardless of direction: $d_{\mathrm{AD}}(x_i, x_j) = d_{\mathrm{AD}}(x_j, x_i)$.
- **Non-negativity**: *AD* uses the Euclidean distance to compute edge weights in the kNN graph, as such all distances are non-negative: $d_{\mathrm{AD}}(x_i, x_j) \geq 0$.
- **Strong triangle inequality**: For any i, j, k, the path between i and k in a tree is contained in the union of the paths between i and j and between j and k. Therefore:

$$d_{\mathrm{AD}}(x_i, x_k) \leq \max\{d_{\mathrm{AD}}(x_i, x_j), d_{\mathrm{AD}}(x_j, x_k)\} \leq d_{\mathrm{AD}}(x_i, x_j) + d_{\mathrm{AD}}(x_j, x_k). \tag{12}$$

Since d_{AD} satisfies the *strong* triangle inequality, it is an *ultrametric* [60].

Cluster centroids, μ_k, are identified as the point (in the graph representation, a vertex) minimizing the squared Euclidean distance to the geometric mean of the cluster, which is equivalent to minimizing the sum of pairwise distances but reduces the cost from $O(N^2D)$ to $O(ND)$. They are constrained to belong to the vertex set, so all distances remain well-defined on the MST.

For $N \leq N_0$ (with threshold $N_0 = 2000$), distances from a source node are computed via the BFS procedure in Algorithm 2, with complexity $O(N)$ per source. For $N > N_0$, a binary lifting table is precomputed in $O(N \log N)$ after MST construction, enabling all N distances from a single source to be computed simultaneously in $O(N \log N)$ via vectorised LCA queries over all nodes in parallel. Query results are cached so that repeated calls for the same source incur only an $O(N)$ copy.

AD constructs an MST by repeatedly visiting vertices and computing distances from the current vertex to all points to select neighbor candidates; this yields a dominant complexity of $O(N^2D)$ for distance computations plus heap/priority-queue overhead of $O(NK \log N)$. Therefore, the total time complexity of *AD* is $O(N^2D + NK \log N)$. The time complexity of a query on the MST is based on breadth-first search (BFS) to find the path between two nodes and to extract its maximum edge, and is in worst-case $O(N)$ per pair. But *AD* constructs the MST with two optimisations: squared-norm reuse, which avoids materialising an (N, D) difference array at each Prim step by computing $\|x_i - x_v\|^2 = \|x_i\|^2 + \|x_v\|^2 - 2x_i \cdot x_v$, and heap pruning via the *best_known* array, which discards stale priority-queue entries and keeps

Algorithm 1 Build Minimum Spanning Tree (MST)

procedure BUILDMST($data, K_{NN}$)
  $visited \leftarrow$ boolean array of size n (all **false**)
  $best_known \leftarrow$ array of size n (all ∞)
  $edges \leftarrow$ empty list
  $pq \leftarrow$ empty min-heap
  $norms_sq[i] \leftarrow \|x_i\|^2$ for all i ▷ precomputed once, $O(ND)$
  $visited[0] \leftarrow$ **true**; $best_known[0] \leftarrow 0$
  $dist_sq \leftarrow norms_sq + norms_sq[0] - 2 \cdot (data \cdot x_0)$
  for each $neighbor$ in K_{NN}-nearest neighbors of point 0 **do**
    if $dist_sq[neighbor] < best_known[neighbor]$ **then**
      $best_known[neighbor] \leftarrow dist_sq[neighbor]$
      PUSH($(dist_sq[neighbor],\ 0,\ neighbor)$) to pq
    end if
  end for
  while $len(edges) < n - 1$ **and** pq is not empty **do**
    $(dist_sq, u, v) \leftarrow$ POP(pq)
    if $visited[v]$ **or** $dist_sq > best_known[v]$ **then** ▷ skip stale entries
      continue
    end if
    $edges$.APPEND($(u,\ v,\ \sqrt{dist_sq})$)
    $visited[v] \leftarrow$ **true**
    $dist_sq_v \leftarrow norms_sq + norms_sq[v] - 2 \cdot (data \cdot x_v)$ ▷ norm reuse
    $dist_sq_v[visited] \leftarrow +\infty$
    $selected \leftarrow$ pick up to K_{NN} nearest unvisited neighbors
    for each $neighbor$ in $selected$ **do**
      if $dist_sq_v[neighbor] < best_known[neighbor]$ **then**
        $best_known[neighbor] \leftarrow dist_sq_v[neighbor]$
        PUSH($(dist_sq_v[neighbor],\ v,\ neighbor)$) to pq
      end if
    end for
  end while
  return $edges$
end procedure

the live heap size $O(N)$. The total time complexity of MST construction is therefore $O(N \cdot K_{NN} \log N)$.

Three CVIs have been extended to use AD instead of the Euclidean distance, specifically Silhouette, Davies-Bouldin, and Calinski-Harabasz. For AD-S, the MST is computed once and distances from all K cluster centroids to all points are obtained in K BFS/lifting traversals, giving a total time complexity of $O(N \cdot K_{NN} \log N + KN)$. For AD-DB, centroid-to-centroid distances are computed via a single call retrieving all K centroid rows at once, replacing the previous $O(K^2)$ individual queries with $O(K)$ traversals; the total complexity is $O(N \cdot K_{NN} \log N + KN)$. AD-CH, similarly

Algorithm 2 Arboris Distance Computation

```
function ArborisDistance(tree, i, j)
    if i = j then
        return 0
    end if
                                  ▷ BFS from i to find path and parent-edge weights
    parent ← {i : None}
    parent_edge_weight ← {}
    queue ← [i]
    while queue is not empty do
        node ← queue.pop(0)
        if node = j then
            break
        end if
        for (nbr, w) in tree.adj[node] do
            if nbr not in parent then
                parent[nbr] ← node
                parent_edge_weight[nbr] ← w
                queue.append(nbr)
            end if
        end for
    end while
                                        ▷ backtrack path j → i, take max edge
    max_w ← 0
    cur ← j
    while parent[cur] is not None do
        max_w ← max(max_w, parent_edge_weight[cur])
        cur ← parent[cur]
    end while
    return max_w
end function
```

to the original CH, computes the between-cluster and within-cluster sums of squares on the MST, with a total time complexity of $O(N \cdot K_{NN} \log N + KN)$.

Arboris index (A) is a separation-to-compactness measure, like many other CVIs. Unlike other CVIs, A index uses cluster-specific MSTs for intra-cluster distances, capturing the internal structure more precisely, while using the global MST for inter-cluster distances to respect overall data connectivity. Through the product over clusters, a balanced view is guaranteed, as the low score of one cluster will affect the whole dataset. A index is a higher-is-better CVI with a $[0, \infty)$ range, where higher values represent well-separated, compact clusters.

The A index computes the MST for each cluster, taking in total $O(N^2 D)$ in the worst case, the same as the full MST, while the centroid-to-centroid distance computations would take $O(K^2 N)$, resulting in a total time complexity of $O(N^2 D + K^2 N)$. Inter-cluster distances are computed via K BFS/lifting traversals on the global

MST, replacing the previous $O(K^2)$ individual queries with $O(K)$ traversals. The total time complexity is $O(N \cdot K_{NN} \log N + KN)$ for the global MST, plus $O(\sum_k N_k \cdot K_{NN} \log N_k)$ for the per-cluster MSTs, where N_k is the size of cluster k.

Our proposed CVI includes the noise handling of DBCV [22] as a multiplication by the percentage of non-noise points in the labels. For each cluster, the index computes the ratio between a minimum inter-cluster term, given by the mean Arboris distance to another cluster, and an intra-cluster term, given by the mean Arboris distance from the cluster centroid to its points. The proposed CVI can be described by the following formula:

$$A \text{ index} = \frac{N - N_{\text{noise}}}{N} \prod_{k=1}^{K} \frac{\min\limits_{k \neq l} \operatorname{mean}\limits_{x \in C_l} d_{\text{AD}}(\mu_k, x)}{\operatorname{mean}\limits_{x \in C_k} d_{\text{AD}}(\mu_k, x)}. \tag{13}$$

2.2 Datasets and labelsets

For the main analyses presented in this work, 44 synthetic clustering benchmark datasets have been used. Table 3 presents the datasets and their characteristics.

For the real-world correlation analysis, we use eight datasets from the UCI data repository. The datasets and their characteristics are presented in Table 4. It has been observed that the UCI datasets do not form sharp clusters [71] and that they are not representative for all problems [71]. It has also been found that DBSCAN provides the lowest performance for many of these datasets [72]. As UCI datasets have been a point of contention in the clustering literature [15, 32, 73], we have also chosen 3 real-world image datasets described in 5. All of these real-world image datasets have been previously used in clustering research [74–77].

For the labelset-based analysis, each synthetic dataset is evaluated using six labelsets. This analysis tests whether a CVI assigns its optimal score to the ground-truth labels rather than to simple alternative partitions. Figure 1 shows the handcrafted labelsets on the ring [69] dataset. These labelsets are created through linear separations of the feature space. We use the following labelsets:

- Ground truth labels (TL);
- First diagonal separated labels (FDL);
- Second diagonal separated labels (SDL);
- Vertical midline separated labels (VL);
- Horizontal midline separated labels (HL);
- Random labels (RL).

2.3 External cluster validity indices

Because external CVIs are used as reference measures in several analyses, we briefly introduce the indices used for comparison. External CVIs measure how closely the clustering labels match the reference labels. Therefore, they have limited applicability to datasets that contain a ground truth [31–33].

The Rand Index (RI) [88] and its extension, the Adjusted Rand Index (ARI) [59, 89, 90], measure clustering performance as the pairwise comparison of two sets of labels. Agreements represent cases where two data points are considered to be in

Table 3: Characteristics of the 44 synthetic benchmark datasets

Name	Characteristics	Size (N)	Dim. (D)	Clusters (K)	Reference
Data1-7	Varying complexities	1000	2	2–5	[37, 61]
aggregation	Irregular shapes, overlap	788	2	7	[56]
compound	Irregular shapes, embedded clusters	399	2	6	[57]
jain	Imbalanced, semi-moons	373	2	2	[62]
pathbased	Irregular shapes, overlapping	300	2	3	[63]
spiral	Irregular shapes	312	2	3	[63]
r15	Overlapping clusters	600	2	15	[64]
flame	Overlap, irregular shapes	240	2	2	[65]
d31	Imbalanced, overlapping	3100	2	31	[64]
unbalance	Imbalanced clusters	6500	2	8	[66]
a1-3	Varying # clusters	3000/5250/7500	2	20/30/50	[55, 67]
s1-4	Increasing overlap	5000	2	15	[68]
parabolic	Semi-moons (interlocking)	1000	2	2	[69]
dense	Imbalanced clusters	200	2	2	[69]
fuzzyx	Overlapping clusters	1000	2	5	[69]
line	Irregular shape	250	2	2	[69]
ring	Concentric circles	1000	2	2	[69]
ring noisy	Concentric circles (noisy)	1000	2	2	[69]
ring outliers	Concentric + outliers	1000	2	2 + 3 outliers	[69]
zigzag	Irregular shape	250	2	3	[69]
zigzag noisy	Irregular (noisy)	300	2	3	[69]
zigzag outliers	Irregular + outliers	280	2	3	[69]
mk1	None	300	2	3	Mateusz Kobyłka (
mk2	Irregular shape	1000	2	2	Mateusz Kobyłka (
smile	Irregular shape	1000	2	6	Przemyslaw Kosew
x1	Imbalanced	120	2	3	Eliza Kaczorek (W
x2	Imbalanced, overlapping	120	2	3	Eliza Kaczorek (W
x3	Imbalanced, overlapping	185	2	4	Eliza Kaczorek (W
trajectories	Irregular shapes	10000	2	4	Michal Maciąg (W
lsun	Irregular shapes	400	2	3	[70]
target	Irregular shapes	770	2	3	[70]
twodiamonds	Irregular shapes	800	2	2	[70]
wingnut	Irregular shapes	1016	2	2	[70]

Table 4: Characteristics of the 8 real-world UCI datasets

Dataset	N	D	K	Reference
ecoli	336	7	8	[78]
glass	214	9	6	[79]
yeast	1484	8	10	[80]
statlog	2310	19	7	[81]
wdbc	569	30	2	[82]
wine	178	13	3	[83]
sonar	208	60	2	[84]
ionosphere	351	34	2	[58]

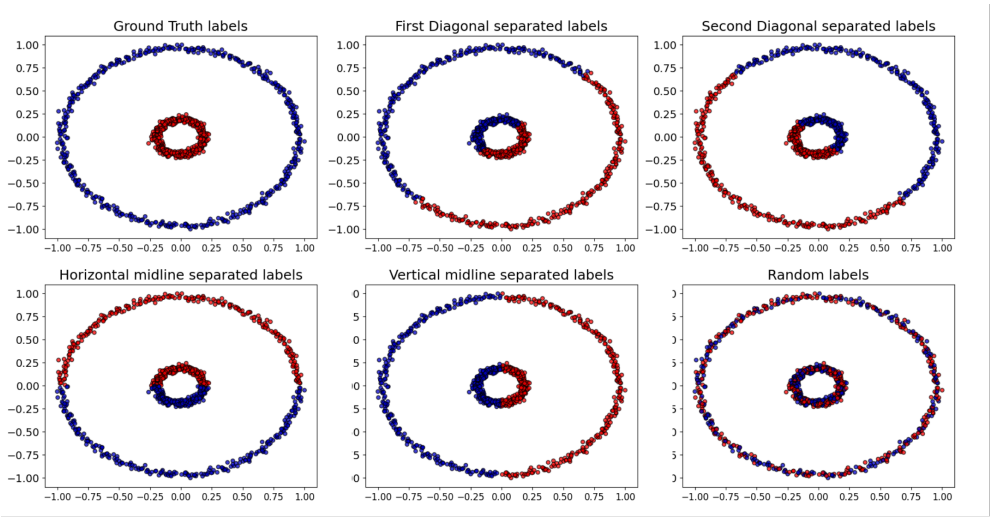


Fig. 1: The six labelsets applied to the ring dataset [69].

the same cluster or in different clusters by both sets of labels, while disagreements represent all other cases. In addition to RI, ARI includes an adjustment for chance agreements. Let N be the total number of data points, a the number of pairs in the same cluster in both partitions, and b the number of pairs in different clusters in both partitions. The indices are defined as:

$$\mathrm{RI} = \frac{\text{agreements}}{\text{agreements} + \text{disagreements}} = \frac{a+b}{\binom{N}{2}}, \qquad \mathrm{ARI} = \frac{\mathrm{RI} - \mathbb{E}[\mathrm{RI}]}{1 - \mathbb{E}[\mathrm{RI}]}, \tag{14}$$

where $\mathbb{E}[\mathrm{RI}]$ represents the expected Rand Index for random cluster assignments and $\max \mathrm{RI} = 1$.

The Mutual Information (MI) and its extension Adjusted Mutual Information (AMI) [90–93] measure the agreement between two partitions through their shared information. Let $U = \{U_1, \ldots, U_r\}$ and $V = \{V_1, \ldots, V_s\}$ be two partitions, let $|U_i|$

Table 5: Characteristics of the 3 real-world image datasets

Dataset	N	D	K	Reference
COIL-20	1440	16384	20	[85]
Olivetti Faces	400	10304	40	[86]
Yale Face Database	165	77760	15	[87]

and $|V_j|$ be the cluster sizes, and N the total number of points. MI is defined as:

$$\mathrm{MI}(U,V) = \sum_{i=1}^{r}\sum_{j=1}^{s} \frac{|U_i \cap V_j|}{N} \log \frac{N \cdot |U_i \cap V_j|}{|U_i| \cdot |V_j|}. \tag{15}$$

The Normalised Mutual Information (NMI) is:

$$\mathrm{NMI}(U,V) = \frac{\mathrm{MI}(U,V)}{\sqrt{H(U)H(V)}}, \tag{16}$$

where $H(U)$ and $H(V)$ are the entropies of the respective partitions:

$$H(U) = -\sum_{i=1}^{r} \frac{|U_i|}{N} \log \frac{|U_i|}{N}, \qquad H(V) = -\sum_{j=1}^{s} \frac{|V_j|}{N} \log \frac{|V_j|}{N}. \tag{17}$$

Finally, the Adjusted Mutual Information (AMI) adjusts MI for chance:

$$\mathrm{AMI}(U,V) = \frac{\mathrm{MI}(U,V) - \mathbb{E}[\mathrm{MI}(U,V)]}{\max(H(U),H(V)) - \mathbb{E}[\mathrm{MI}(U,V)]}, \tag{18}$$

where $\mathbb{E}[\mathrm{MI}(U,V)]$ is the expected mutual information under a random model of cluster assignments.

These external measures provide reference scores for comparing clustering labelsets, while the next subsection defines the dataset-level quantities used to characterize imbalance and overlap.

2.4 Imbalance and overlap measures

Two important factors affecting clustering and CVI behaviour are *imbalance* and *overlap*. To measure the amount of imbalance and overlap in a dataset, the imbalance ratio (IR) and overlap ratio (OR) were used.

The IR is computed as the ratio between the count of the largest cluster and the count of the smallest cluster:

$$\mathrm{IR} = \frac{\max_{1\le k\le K} n_k}{\min_{1\le k\le K} n_k}. \tag{19}$$

The OR first computes the centroid of each cluster. For each point, it compares the distance to its own centroid with the distance to the nearest other centroid. If the nearest other centroid is within a configurable slack parameter $\lambda = 1.2$ of own-centroid distance, the point is counted as overlapping, as shown in Algorithm 3. Finally, the function returns the fraction of points that are overlapping.

These two quantities are used in the first result analyses to relate CVI failures to controlled changes in imbalance and overlap.

Algorithm 3 Overlap Ratio

```
function OVERLAP_RATIO(X, labels, slack = 1.2)
    unique_labels = ordered list of distinct labels
    centres = empty list
    for each label IN unique_labels do
        points = subset of X with that label
        centre = mean vector of points
        append centre to centres
    end for
    N = number of rows in X
    overlapping_count = 0
    for i from 1 to N do
        p = row i of X
        own_idx = index of label of p in unique_labels
        dist_to_own = EuclideanDistance(p, centres[own_idx])
        dist_to_nearest_other = +∞
        for j from 1 to length(centres) do
            if j == own_idx then
            end if
            d = EuclideanDistance(p, centres[j])
            if d < dist_to_nearest_other then
                dist_to_nearest_other = d
            end if
        end for
        if dist_to_nearest_other ≤ slack × dist_to_own then
            overlapping_count = overlapping_count + 1
        end if
    end for
    overlap_rate = overlapping_count/N
    return overlap_rate
end function
```

3 Results

3.1 Imbalance and overlap effects on internal CVIs

We first illustrate how imbalance and overlap can affect the behaviour of a representative internal CVI, using the Silhouette index (S).

We first use a simple three-cluster dataset and vary the amount of imbalance by changing the size of one cluster (green). Figure 2 shows that the Imbalance Ratio tracks the intended increase in imbalance as the minority cluster becomes smaller.

We can see that the Silhouette index is able to correctly evaluate the true labels with a higher score for most cases of imbalance. However, for heavily imbalanced datasets, where close sparse clusters are present, it may misrank the true labels, as shown in Figure 2 (bottom row) for an Imbalance Ratio of 10.

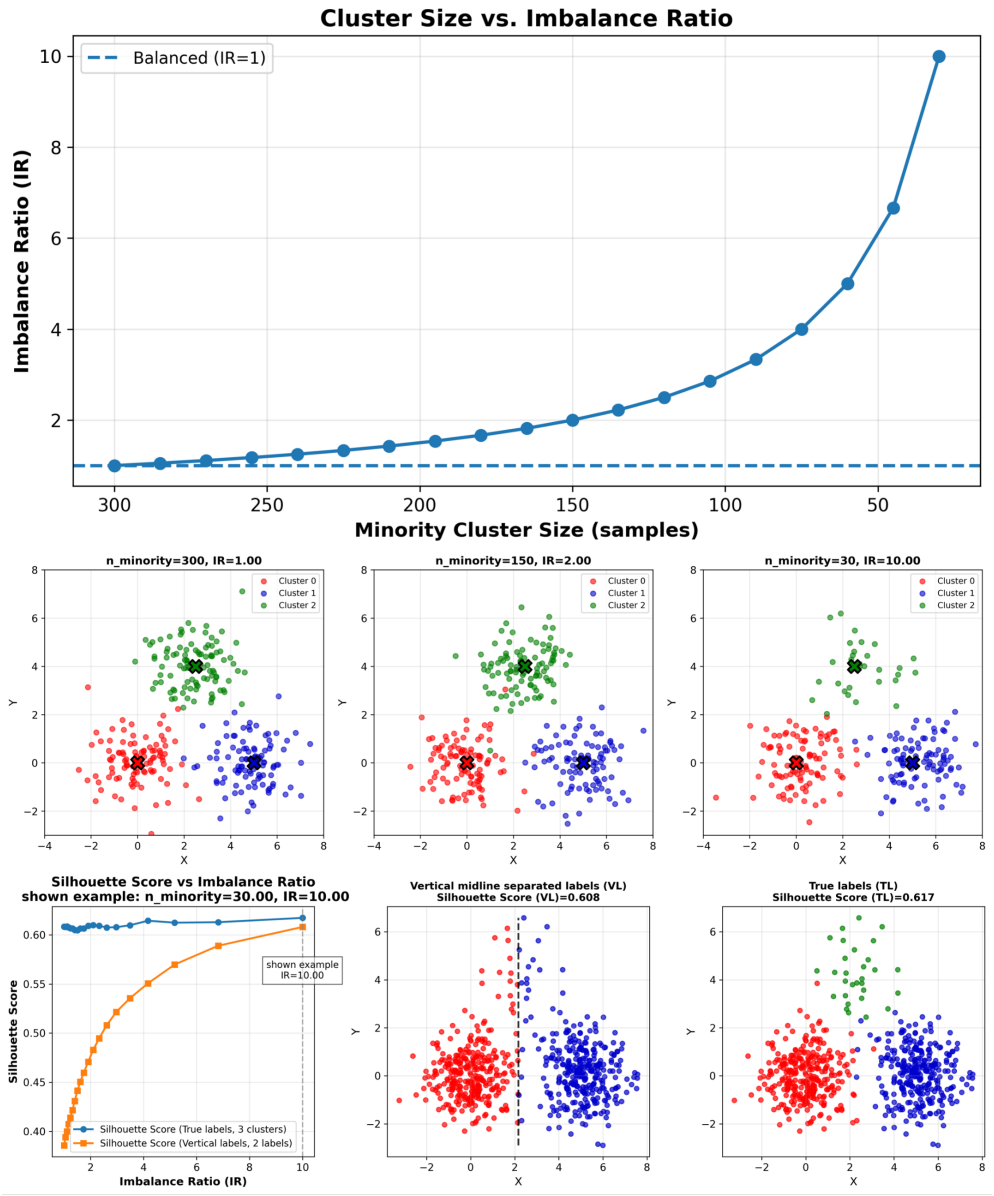


Fig. 2: Impact of imbalance on the Silhouette index (S) internal CVI.

The analysis continues with an analogous scenario for increasing levels of overlap. Figure 3, middle row, shows that the Overlap Ratio measure tracks the intended increase in overlap, with some small variations due to regeneration of clusters. In contrast to imbalance, even a small amount of overlap can result in an erroneous evaluation for CVIs, as shown in Figure 3, bottom row. Therefore, there is a point

at which, even when a clustering algorithm still recovers a reasonable partition, a Silhouette analysis would be unable to correctly determine the number of clusters.

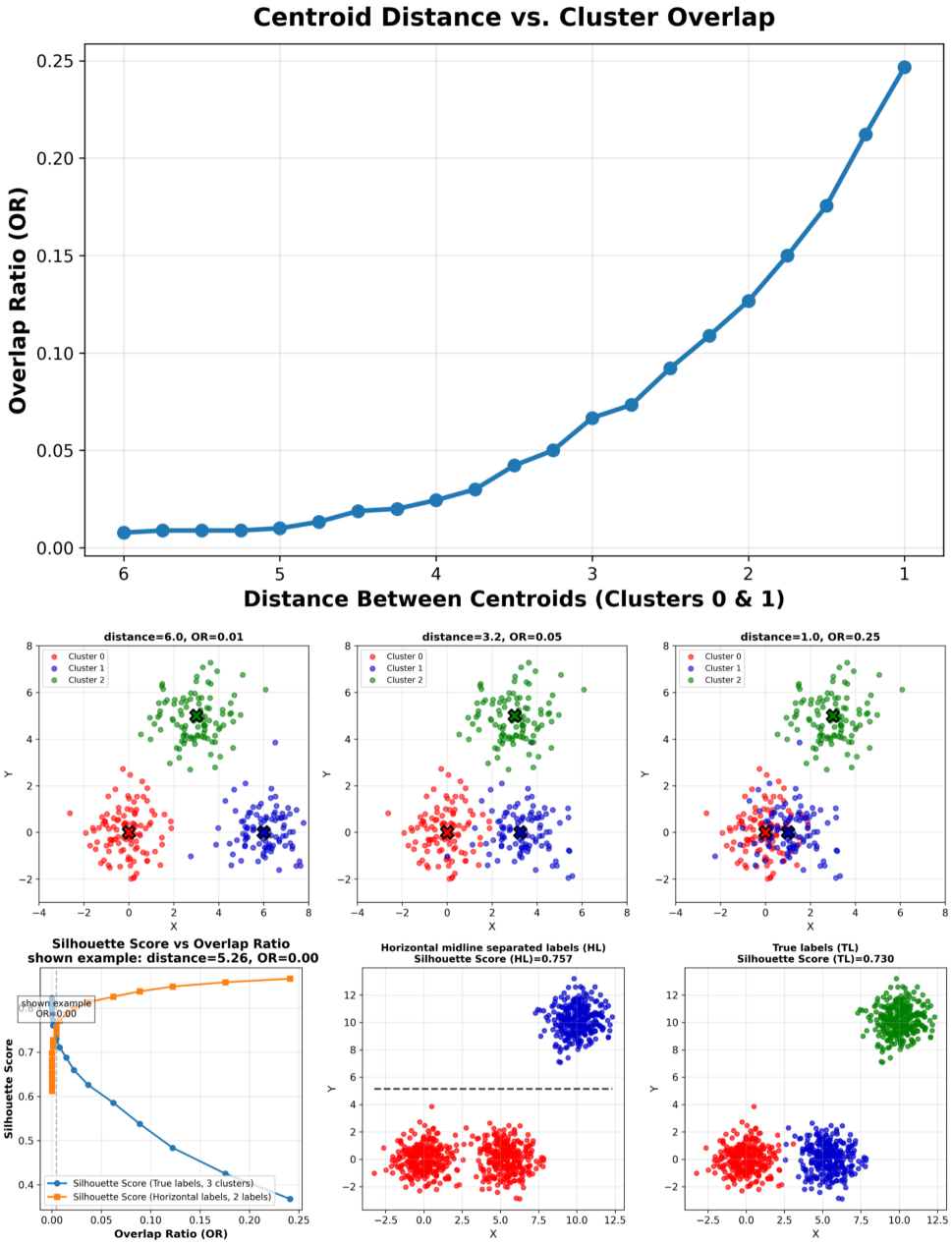


Fig. 3: Impact of overlap on the Silhouette index (S) internal CVI

To show that this behaviour is not limited to handcrafted labelsets, we repeat the comparison using K-Means solutions with $k = 2$ and $k = 3$ under the same overlap setting. As shown in Figure 4, the CVI can again prefer an incorrect or suboptimal partition.

These controlled examples motivate the broader labelset-based and algorithm-generated analyses that follow.

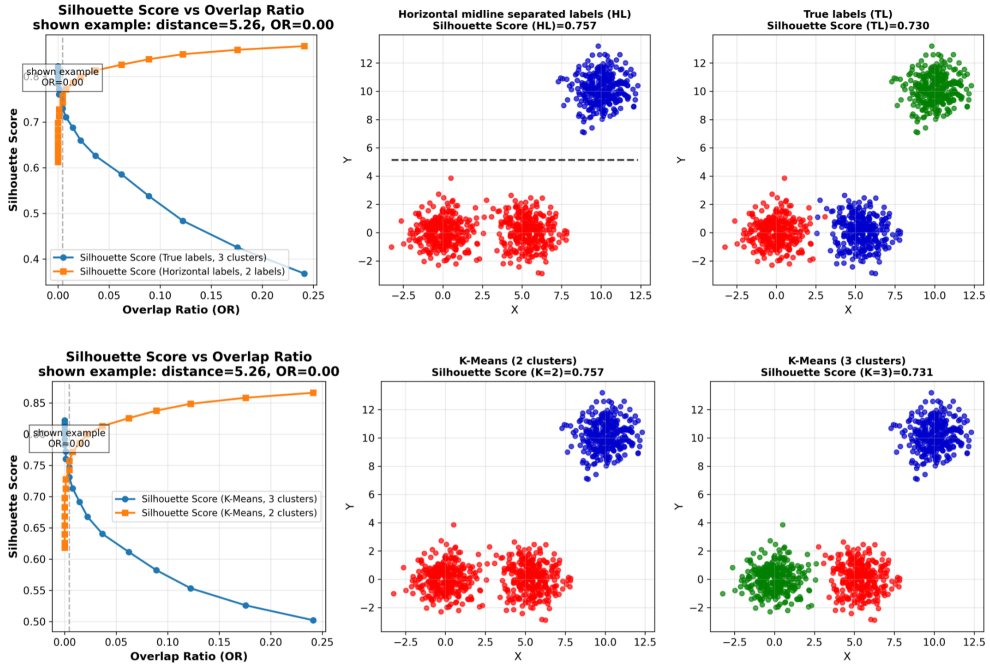


Fig. 4: Comparison between handcrafted labelsets and K-Means solutions for different values of k.

3.2 Estimating the number of clusters using internal CVIs

We next examine the use of internal CVIs for parameter selection, focusing on the choice of k in K-Means. The ground-truth labels are used as a reference to test whether the Silhouette index (S), a commonly used CVI for this task, assigns its optimal score to the ground-truth labelset.

To test whether this behaviour also appears for algorithm-generated labelsets, we compare the ground-truth labels with K-Means solutions for $k = 2$ and $k = 3$. Figure 5 shows that the ground truth has a substantially lower validation score than either two or three clusters of K-Means, which will be an issue for any clustering algorithm, as the Silhouette index prefers the K-Means solutions over the ground-truth labelset.

This indicates that, when searching for parameters based on the S values, even if a clustering algorithm can potentially find a clustering solution matching the true

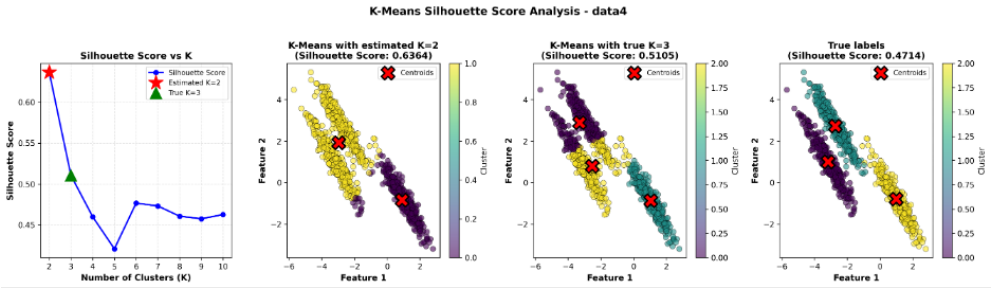


Fig. 5: Estimating the k parameter in K-Means using the Silhouette index (S).

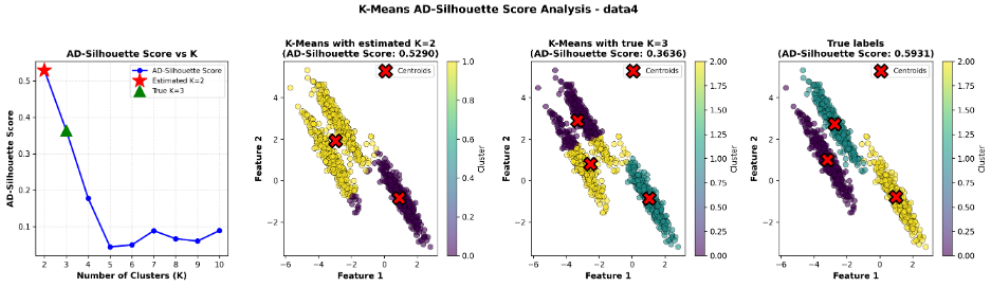


Fig. 6: Estimating the k parameter in K-Means using the proposed AD-Silhouette index (AD-S).

labels, it will reject it, due to the low S score. This phenomenon can lead to suboptimal clustering results.

In the analogous AD-Silhouette analysis shown in Figure 6, a similar trend is found where K-Means with $k = 2$ still obtains a better result; however, the true labels obtain higher scores than the K-Means solutions with $k \in \{2, 3\}$.

3.3 Comparison of classical CVIs with Edging- and Arboris-Distance extensions

In this first analysis, the performance of the Arboris Distance-extended CVIs is shown in comparison to the original CVIs and the Edging Distance-based extensions on the ring [69] dataset. Table 6 shows that, for a dataset with irregular cluster shapes (concentric circles for the ring dataset), both the AD- and ED- extensions can more accurately assign the optimal score to the true labels (TL), whilst the original S, DB, and CH result in erroneous assignment (indicated by *) of better scores for other labelsets. Moreover, the AD extensions offer a clearer separation between the TL and the other labels and offer a significant speedup to execution time compared to their ED-based equivalents.

To demonstrate that this is not an exception, the performances of these CVIs on the 44 synthetic benchmark datasets have been aggregated. Table 7 reports the number of correct evaluations, where the TL obtain the optimal scores (highest/lowest,

Table 6: Comparison of original CVIs vs Edging Distance extensions vs Arboris Distance extensions on the ring dataset.

	TL	FDL	SDL	HL	VL	RL	Time (s)
S (↑)	0.268	0.254	0.255	0.254	0.257	0	0.014
ED-S (↑)	0.879	-0.154	-0.194	-0.184	-0.171	-0.209	2.820
AD-S (↑)	0.94	0.003	0.003	0.008	0.013	0.006	0.055
DB (↓)	659.861	1.424*	1.420*	1.427*	1.414*	29.106*	0.001
ED-DB (↓)	0.286	18.018	21.07	16.424	18.393	20.062	1.040
AD-DB (↓)	0.12	17.37	22.453	15.367	17.197	37.897	0.054
CH (↑)	0.002	388.871*	389.559*	386.238*	394.016*	0.815*	0.000
ED-CH (↑)	4098.263	102.526	105.927	130.234	99.984	73.004	1.145
AD-CH (↑)	8309.4	97.364	84.687	105.225	98.268	81.484	0.080

depending on the CVIs) in comparison to the other labelsets and the number of errors where one out of the 5 labelsets obtains a better score than the TL. It is clear that the AD extensions improve upon the results of the ED extensions in all cases, and that for both the S and DB, they substantially improve the overall performance of the CVIs with a higher number of correct evaluations and a lower number of errors. However, in the case of the CH, the extensions do not improve as strongly. This decrease for the CH is due to the A1-3 [55, 67] and S1-4 [68] datasets that have a high number of clusters and overlap with regular shapes, where the CH fares better (though this is not the case for datasets with irregular shapes).

Table 7: Aggregated comparison of original CVIs vs ED extensions vs AD extensions on the 44 syntheti datasets.

	S (↑)	ED-S (↑)	AD-S (↑)	DB (↓)	ED-DB (↓)	AD-DB (↓)	CH (↑)	ED-CH (↑
Correct evaluations (out of 44)	23	34	37	24	22	31	23	2
Errors (out of 220)	63	21	15	70	49	31	68	8

We have chosen to exclude the ED extensions from the rest of the analyses due to the higher execution times and lower performance relative to the AD extensions, as shown by the analyses in this subsection.

3.3.1 How dependent is the Arboris Distance on the number of neighbors

One question that easily arises from the proposed approach is how dependent it is on the K_{NN} parameter. Due to the use of the MST on the kNN graph, increasing the number of neighbors is expected to induce few changes, because the MST retains only the lowest-weight connectivity structure. Figure 7 presents an empirical analysis of the parameter for the new CVIs introduced on the simple D1 dataset, showing the

values obtained for each CVI by varying the K_{NN} parameter in the $[3, 50]$ range. The impact is subsequently small on all CVIs with no differences in the values obtained for values greater than 9, and minor variations for smaller values.

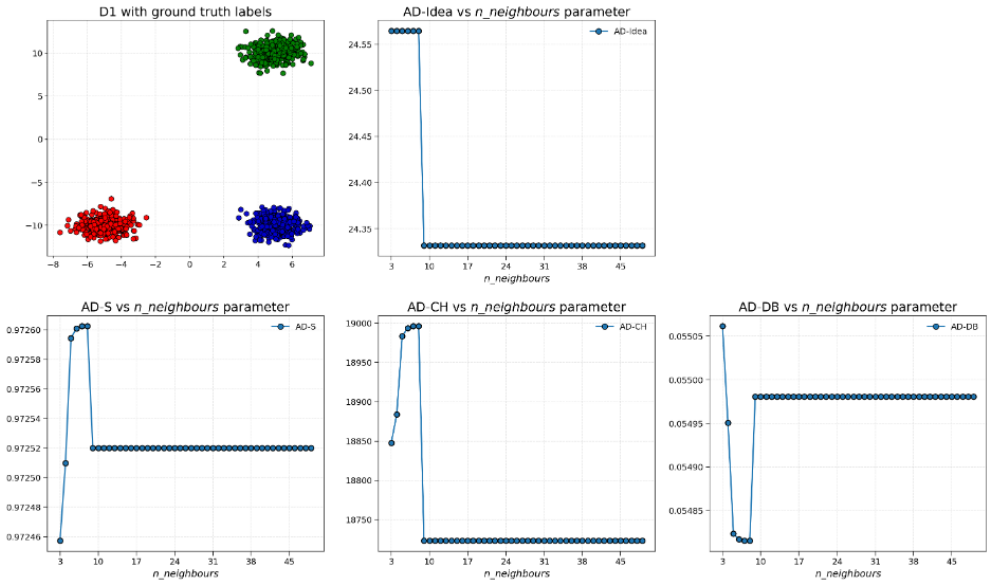


Fig. 7: The impact of the K_{NN} parameter on the proposed CVIs in the $[3, 50]$ range.

3.4 Aggregated CVI ranking results

The aggregated analysis in Table 8 includes 29 baseline internal CVIs together with the proposed AD-based CVIs across 44 synthetic benchmark datasets. We count a *correct* evaluation when TL receives the optimal CVI score among the six labelsets, and an *error* when at least one of the five alternative labelsets receives a better score than TL. This captures both how often a CVI selects TL and how often it misranks alternative labelsets above TL. Our results (the first two columns of Table 8) indicate that the CDbw, DBCV, and the A index are the best-performing CVIs.

This initial analysis was extended with a similar analysis made for a large set of parametrizations (50 to 100) for each dataset for a given clustering algorithm. In the case of algorithm-generated labelsets, the correct count was increased when the CVI's optimal score (either highest or lowest) was obtained on the same parametrization as the highest value of ARI; thus, the maximum count is equal to the number of datasets, which is 44. The error count was increased when a parametrization resulted in a more favourable CVI score than the one given by the parametrization with the highest value of ARI; thus, the maximum count is equal to the number of datasets multiplied by the number of predefined parametrizations for a clustering algorithm. This second analysis, which includes six different clustering algorithms with different approaches

Table 8: Overall analysis. Bolded values indicate the best values by column.

	Labelsets		Agglomerative		Clustering		DBSCAN		HDBSCAN		K-Means		MeanShift	
	Cor.	Err.	Cor.	Err.	Cor.	Err.	Cor.	Err.	Cor.	Err.	Cor.	Err.	Cor.	Err.
cSIL	24	65	8	1273	8	730	10	2207	19	1014	13	418	11	1027
GD43	23	73	8	1103	9	693	9	2255	16	918	9	496	10	941
GD53	1	180	5	1178	8	736	6	2395	16	1194	7	505	9	831
PS	26	76	8	852	10	707	8	2312	18	894	13	426	12	610
rCIP	11	116	4	1197	8	739	7	2315	18	852	6	378	4	849
WB	26	61	11	996	9	691	11	2159	17	996	12	456	9	1035
XB	22	76	11	1082	8	732	8	2303	21	774	11	333	8	934
SSE	26	58	1	1154	10	284	17	982	7	1386	7	610	1	1256
RS	24	68	6	1159	8	716	10	2204	16	966	12	468	10	893
DH	30	39	0	1448	1	447	11	810	9	1290	5	577	1	1269
B	27	53	1	1261	2	492	12	1073	8	1266	6	584	1	1238
BH	32	30	1	1148	0	505	12	811	7	1398	7	614	0	1307
$\mathcal{D}$	14	106	3	1353	2	486	11	1219	18	846	5	419	6	1017
H	10	113	7	1335	8	731	8	2303	16	1164	8	404	6	914
DBCV	41	3	9	699	6	345	12	876	15	936	10	371	16	383
I	21	69	10	1049	0	458	13	755	19	774	14	282	9	954
C	26	59	6	1050	16	190	19	733	22	714	11	457	10	950
CDbw	40	5	2	1011	2	460	10	832	10	1230	7	459	1	1185
VIASCKDE	21	62	8	1282	13	347	17	1079	18	828	11	453	12	747
COP	25	67	3	986	11	311	**22**	660	12	1038	11	450	3	874
Sym	28	43	11	930	12	253	16	836	17	954	9	465	10	765
CS	24	57	4	1414	5	445	13	850	20	1002	6	544	7	1057
SF	11	109	7	1126	6	460	13	1256	18	894	11	357	7	627
SD	30	42	1	1181	4	442	14	700	7	1380	6	562	1	1248
SDbw	29	41	5	1148	6	381	13	705	16	1050	8	548	7	1044
XB*	18	87	9	1162	7	380	11	1300	18	960	9	400	9	905
S	23	63	13	946	19	190	21	758	22	732	**15**	288	11	794
DB	24	70	6	1281	7	456	14	922	21	882	13	370	9	951
CH	23	68	10	976	16	203	21	662	18	774	12	431	8	943
AD-S	37	15	**17**	555	**23**	**178**	**22**	555	22	**606**	13	355	**22**	260
AD-DB	31	31	11	794	13	358	15	750	19	762	14	**273**	20	**235**
AD-CH	22	78	16	497	18	221	20	905	22	828	12	340	13	411
A	**44**	**0**	14	**345**	5	276	19	**409**	**25**	624	**15**	395	19	307

to clustering (such as partitional, density-based, hierarchical), supports our claim that the proposed CVIs provide more consistent ranking behaviour due to the fact that both the number of correct evaluations is high and the number of errors is low. A low number of errors indicates that across multiple datasets, the CVI consistently evaluates parametrizations with "worse" scores than the "best" parametrization.

Our proposed CVIs consistently assign more favourable scores to the highest ARI parametrizations, given by the higher number of correct evaluations across all clustering algorithms. Even though DBCV and CDbw were capable of assigning the highest score to the ground truth, the second analysis reveals that they are not entirely capable of assigning the highest score to the best clustering, which is an issue when they differ.

3.5 Real-world correlation analysis

Following previous work on DBCV [22], we use Pearson correlations between ARI and each CVI as an additional, imperfect, real-world validation strategy [22, 94, 95].

We consider 8 real datasets, multiple clustering algorithms and multiple parametrizations: DBSCAN, HDBSCAN, MeanShift, AgglomerativeClustering, SpectralClustering, and K-Means. The current analysis can be separated into the following steps:

1. For a given dataset:
   (a) For a given clustering algorithm:
      (i) Run grid search (50-100 parametrizations) and save all clustering labelsets;
2. For each dataset:
   (a) For each CVI:
      (i) Compute ARI and CVI given for all the clustering labels regardless of algorithm or parametrization;
      (ii) Compute Pearson correlation between ARI and CVI.

The results of this analysis are presented in Table 9. An important thing to consider when analyzing correlation values is the range of the CVI. For lower-is-better CVIs, negative correlations are relevant, while positive correlations mean the metric cannot identify the performance of the clustering correctly; thus, for lower-is-better CVIs, all correlations have been negated. The analogous situation for higher-is-better CVIs, negative correlations indicate disagreement with ARI. CDbw results are not presented as their execution time was too high.

In our previous analysis, the AD-based CVIs obtained the best performance. On the real-world correlation analysis, however, the best performing CVI is CH for almost all datasets. In fact, it can be seen that the performances are reversed for a subset of metrics when compared with the previous analysis on synthetic benchmark datasets. Specifically, WB, SSE, BH, and COP also obtain high correlations on most real datasets, although they performed poorly for synthetic datasets.

The proposed correlation analysis alone is insufficient and can be misleading. As such, we have complemented the correlation analysis with the same count analysis used above: a correct evaluation occurs when the CVI optimum matches the ARI-optimal parameterization, and an error occurs when another parameterization receives a more favourable CVI score than the ARI-optimal one.

A subset of CVI values are reported as `NaN` when the computation failed. For example, CDbw failed on some high-dimensional datasets because the convex hull required for its computation could not be reliably constructed.

3.6 Real-world image dataset analysis

The UCI datasets are a point of contention in the clustering literature because their class labels overlap substantially and may not correspond to a well-separated cluster structure [15, 32, 73]. To complement this analysis, we use three real-world image datasets previously used in clustering research, providing a high-dimensional setting

Table 9: Correlation analysis by real datasets across various clustering algorithms.

	ecoli	glass	yeast	statlog	wdbc	wine	ionosphere	sonar	Correct (out of 48)	Errors
cSIL	0.226	-0.173	-0.165	-0.715	-0.372	-0.455	-0.269	-0.591	8	1176
GD43	0.437	-0.066	-0.608	-0.703	0.498	-0.089	-0.049	-0.600	5	1237
GD53	0.103	0.068	-0.374	0.120	0.661	0.335	-0.124	0.466	3	1048
PS	-0.069	-0.123	0.242	0.579	0.574	0.543	0.732	0.618	5	909
rCIP	0.288	-0.285	-0.422	-0.547	-0.355	-0.272	-0.558	-0.587	5	1180
WB	0.135	0.222	0.459	0.663	0.864	0.873	0.828	0.895	8	799
XB	0.573	0.020	-0.440	-0.541	0.327	-0.021	-0.213	-0.289	5	1178
SSE	0.376	-0.095	0.597	0.637	0.730	0.567	0.777	0.798	9	819
RS	0.192	-0.278	-0.584	-0.445	0.067	0.116	-0.498	-0.271	4	1166
DH	0.244	-0.355	-0.051	-0.225	-0.304	-0.455	0.271	-0.091	9	1165
B	0.607	-0.211	-0.344	-0.444	0.382	-0.406	0.277	-0.009	7	989
BH	0.246	0.262	0.378	0.277	0.313	0.010	0.130	0.356	9	909
$\mathcal{D}$	-0.001	0.056	-0.667	-0.525	-0.389	-0.551	-0.514	-0.570	5	1090
H	0.288	-0.298	-0.685	-0.545	-0.269	-0.243	-0.499	-0.477	4	1176
DBCV	-0.336	0.231	-0.319	-0.660	-0.293	-0.748	-0.487	-0.717	6	1065
I	-0.015	0.096	-0.296	-0.521	0.388	-0.340	-0.514	-0.002	7	896
C	0.388	0.522	0.302	-0.299	0.553	0.156	0.908	-0.249	8	867
CDbw	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	-	-
VIASCKDE	0.403	-0.034	-0.077	-0.717	0.259	0.266	0.530	0.249	13	1057
COP	-0.001	-0.410	0.358	0.434	0.835	0.606	0.398	0.117	11	642
Sym	0.627	0.011	-0.162	-0.574	0.766	0.057	0.465	0.440	9	755
CS	0.284	-0.306	-0.369	-0.624	-0.454	-0.223	0.124	-0.458	11	949
SF	-0.340	-0.159	-0.501	-0.258	-0.129	0.004	-0.328	-0.571	7	914
SD	0.138	-0.370	-0.134	-0.311	-0.127	-0.283	0.299	-0.216	10	1090
SDbw	0.196	-0.617	-0.445	-1.000	-0.247	0.021	0.036	-0.165	8	1461
XB*	0.302	-0.201	-0.403	-0.601	0.377	-0.056	0.170	-0.323	7	1050
S	0.664	0.417	0.057	-0.587	0.547	-0.071	0.644	-0.232	9	766
DB	0.297	-0.443	-0.356	-0.611	0.037	-0.193	0.124	-0.281	8	984
CH	0.529	0.230	0.437	0.560	0.794	0.862	0.689	0.893	11	535
AD-S	-0.040	0.209	-0.277	-0.684	-0.336	-0.240	-0.504	-0.663	9	1005
AD-DB	0.244	-0.270	-0.505	-0.680	-0.442	-0.405	-0.549	-0.428	9	977
AD-CH	0.226	0.541	0.273	0.297	0.521	0.785	0.032	0.564	10	587
AD-idea	-0.129	-0.367	-0.053	-0.339	0.134	0.090	-0.223	-0.115	9	843

with visually meaningful object or identity labels [74–77]. The three datasets, COIL-20 [85], Olivetti Faces [86], and the Yale Face Database [87], contain varying image sizes that have been resized to 64×64 and vectorized into 4096 features due to computational considerations. In contrast to the previous analysis, correlations between the internal CVIs and the ARI are computed separately for each clustering algorithm, pooling all generated labelsets across the image datasets. Each clustering algorithm has a tendency towards the clusters they create; as such, computing the correlation in this way provides a different perspective. Each clustering algorithm has been run on multiple parametrizations, and all labelsets have been used in this computation, approximately 500 in total. DBSCAN and MeanShift have been removed from this analysis as they did not identify meaningful clusters, i.e., all their generated labelsets

Table 10: The best ARI scores obtained for each clustering algorithm on the real-world image datasets.

	KMeans	DBSCAN	MeanShift	SpectralClustering	AgglomerativeClustering	HDBSCAN
COIL-20	0.610	0.000	0.000	0.727	0.714	0.771
Olivetti Faces	0.616	0.000	0.000	0.763	0.741	0.166
Yale Face Database	0.456	0.000	0.000	0.483	0.425	0.342

received an ARI score of 0.0. The best ARI results obtained by each clustering algorithm are shown in Table 10.

The results of this analysis are presented in Table 11. For K-Means, none of the CVIs manage to achieve a high correlation. This is likely due to the high variation in clustering results depending on the parametrization with DBCV obtaining the highest correlation of 0. For AgglomerativeClustering, it is the $\mathcal{D}$ index followed closely by DBCV that obtain the highest correlations with ARI. For SpectralClustering and HDBSCAN, it is the proposed Arboris Index that obtains the highest correlation across all CVIs. It is notable that SpectralClustering is able to obtain the highest ARI scores for these datasets as shown in Table 10.

When comparing the correlation analysis with the complementary counting analysis of Table 11, the two analyses emphasize different aspects of CVI behaviour. Correlations tell us that the proposed A index has the highest overall performance, whilst the count analysis indicates that DBCV and AD-S have the highest number of correct evaluations, followed by COP with A being the fourth. Thus, we can conclude that, while both analyses offer a high-level perspective about potential performance, they do not actually give a clear picture. As such, we have selected and aggregated data that was used in the correlation and count analyses to view how these 4 CVIs behave on the real image datasets. To visualize these trends, we sort the ARI scores for all parametrizations of each clustering algorithm. The same ordering is then applied to the corresponding CVI scores. Figure 8 shows a high-level 2×2 grid showing this analysis for each clustering algorithm. Each panel contains a 5×3 grid, where rows represent the CVIs and the columns the datasets. In the panels, the first row (in gray) is always the sorted-by-score ARI, while the other CVIs received the same ordering. Figure 8 shows an interesting trend for DBCV and COP, while being evaluated with a high correct count their scores are quite indistinguishable across parametrizations. This is important because it means that almost the same scores will be obtained if the clustering is arbitrary (ARI $= 0$) or if it is a correct one (ARI $= 1$). AD-S shows an increasing trend similar to ARI. Because the A index has no upper bound, its values remain small for low-ARI parameterizations, but increase sharply for high-ARI parameterizations (overshadowing neighboring values). We consider this a valuable characteristic: it suggests that the proposed A index may facilitate parameter selection compared with other CVIs, because high-quality parameterizations are separated by much larger score differences. For example, distinguishing between DBCV scores of 0.99 and 0.98 is less direct than distinguishing between A index values of 100 and 2×10^6.

Table 11: Correlation analysis by various clustering algorithms across 3 real image datasets.

	AgglomerativeClustering	HDBSCAN	KMeans	SpectralClustering	Correct (out of 12)	Errors
cSIL	-0.854	NaN	-0.579	-0.173	0	584
GD43	0.019	NaN	0.864	0.334	2	431
GD53	0.560	NaN	-0.273	0.244	1	485
PS	0.487	NaN	0.100	0.414	2	350
rCIP	NaN	NaN	NaN	NaN	-	-
WB	0.085	NaN	-0.907	-0.134	1	387
XB	-0.130	NaN	0.711	0.171	1	447
SSE	0.069	-0.590	-0.432	-0.254	4	179
RS	-0.834	NaN	-0.889	-0.451	1	516
DH	-0.370	-0.665	0.389	0.074	4	347
B	0.112	-0.570	-0.693	0.005	3	293
BH	0.542	-0.176	0.225	0.203	5	179
$\mathcal{D}$	0.151	-0.701	0.289	0.383	2	414
H	-0.849	-0.882	-0.904	-0.391	1	531
DBCV	**0.714**	0.654	0.063	-0.098	8	119
I	-0.219	-0.828	-0.918	-0.352	0	410
C	0.247	0.469	-0.404	0.318	1	305
CDbw	NaN	NaN	NaN	NaN	-	-
VIASCKDE	NaN	NaN	NaN	NaN	0	881
COP	0.667	0.760	-0.329	0.360	6	191
Sym	0.054	-0.728	-0.611	-0.278	0	327
CS	-0.830	0.523	-0.493	-0.193	4	360
SF	NaN	-0.354	NaN	NaN	0	860
SD	0.048	-0.737	0.382	0.179	3	407
SDbw	NaN	NaN	NaN	NaN	0	881
XB*	-0.305	-0.359	**0.864**	0.223	2	386
S	0.324	0.588	-0.811	0.165	4	178
DB	-0.790	0.356	-0.529	0.044	2	376
CH	0.501	0.824	-0.389	0.231	2	251
AD-S	0.093	0.815	-0.461	0.283	8	97
AD-DB	-0.734	0.611	-0.500	-0.193	5	312
AD-CH	0.456	0.565	-0.389	0.159	1	333
A	0.391	**0.920**	0.582	**0.875**	5	89

4 Conclusions

This work presented a systematic evaluation of the limitations of CVIs for clustering validation across a diverse assortment of datasets and evaluation scenarios. The results show that most CVIs often fail to reliably identify the reference labelset (i.e., externally best-supported), particularly when datasets contain irregular cluster shapes, imbalance or overlap. Our analyses show that internal and external CVIs may reach breaking points, as overlap or imbalance increases by showcasing the Silhouette index and Adjusted Rand Index. Internal CVIs are influenced less by imbalance, yet tend to be sensitive to overlap, resulting in erroneous performance evaluations. Internal CVIs are sensitive to the number of clusters, perhaps too sensitive to use them as validation for the choice of this parameter. However, external CVIs are invariant to overlap (due

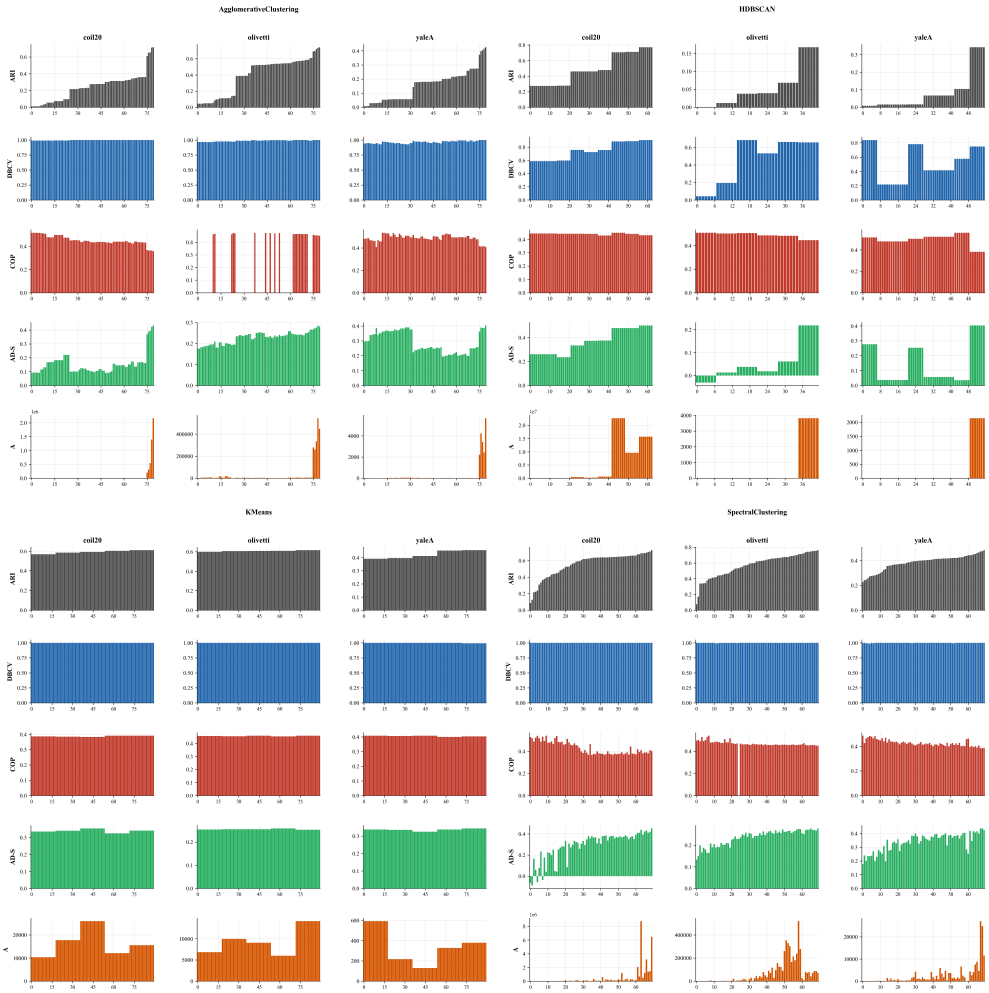


Fig. 8: Score comparisons across all parametrizations, ordered by ARI score for DBCV, COP, AD-S and A indices (increasing trends mean that the metric has a higher score when ARI also has a higher score).

to their matching approach), but tend to be sensitive to cluster imbalance, resulting in misleading evaluations.

Our findings reveal that the majority of the 29 CVIs evaluated frequently assign more favourable scores to incorrect labelsets rather than the ground truth labels or even the ARI-optimal algorithm-generated labelsets. This fundamental flaw diminishes their utility in the validation of clustering performance and parameter selection, such as estimating the number of clusters. This flaw is consistent with the known bias of many CVIs toward globular and well-separated clusters [15, 55].

The proposed Arboris Distance addresses these limitations through a graph-based approach, allowing for the inclusion of the data structure in the distance computation. By computing distances along a minimum spanning tree, AD adapts to irregular shapes while maintaining metric properties. The AD-based CVIs show improved ranking performance (i.e., higher number of correct evaluations and a lower number of errors) in comparison to other CVIs. On the one hand, the newly developed A index achieves perfect performance on the handcrafted labelsets analysis with zero errors and is among the best performing for the clustering labelsets. On the other hand, the real-dataset analyses provide a more nuanced picture. This is expected, since CVI behaviour is inherently data dependent. Furthermore, a discussion has been emerging in the literature about the use of UCI datasets in clustering [15, 32]. Prior work has shown that classes in these datasets overlap too strongly to clearly distinguish separate clusters [73], as the matching between clusters and labels often breaks down [32]. Moreover, it has been pointed out that class information is related to properties of the individual objects [73], rather than inter-object relationships as clusters are. This raises the question of whether these datasets are fit to be used in the validation of clustering. This motivates the use of synthetic benchmarks [55], which has been considered. Additionally, our analysis of real-world image datasets shows that the proposed Arboris index has the highest correlation with ARI for 2 out of 4 clustering algorithms. Furthermore, our selected-and-aggregated data analysis shows that the A index obtains high scores when ARI does, more so than other CVIs.

Nevertheless, no single CVI, including the proposed ones, can achieve perfect validation across all datasets and scenarios. This is confirmed by our analysis of correct evaluations and errors on the labels obtained by clustering algorithms; this indicates that there are cases in which every CVI fails. The choice of CVI will likely remain dependent upon the characteristics of the data, and domain knowledge remains one of the most important factors for obtaining optimal results. In conclusion, while this work reveals several serious limitations in the current approaches to clustering validation, it also provides more reliable CVIs. The Arboris Distance and its associated CVIs should therefore be viewed as a step toward more reliable clustering validation, not as a universal solution.

Declarations

Some journals require declarations to be submitted in a standardised format. Please check the Instructions for Authors of the journal to which you are submitting to see if you need to complete this section. If yes, your manuscript must contain the following sections under the heading 'Declarations':

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- Conflict of interest/Competing interests (check journal-specific guidelines for which heading to use)
- Ethics approval and consent to participate
- Consent for publication
- Data availability: All data is publicly available.
- Materials availability
- Code availability: The code was written in Python and is available at: https://github.com/ArdeleanRichard/Clustering-Validity
- Author contribution

Appendix A Mathematical definitions of CVIs

This section provides the mathematical formulations of the CVIs used in this work. The notation follows Table 1; unless otherwise stated, $d(\cdot,\cdot)$ denotes the distance used by the corresponding index.

Compactness and separation

Many internal CVIs combine within-cluster compactness and between-cluster separation. *Compactness* is commonly measured through the within-cluster sum of squares:

$$\mathrm{SSE}(C_k) = \sum_{x \in C_k} d(x, \mu_k)^2, \qquad \mathrm{SSE} = \mathrm{SSW} = \mathrm{WSS} = \sum_{k=1}^{K} \mathrm{SSE}(C_k) = \sum_{k=1}^{K} \sum_{x \in C_k} d(x, \mu_k)^2. \tag{A1}$$

Separation is commonly measured through the between-cluster sum of squares:

$$\mathrm{SSB} = \mathrm{BSS} = \sum_{k=1}^{K} n_k \cdot d(\mu_k, \mu_{\mathrm{data}})^2. \tag{A2}$$

Davies-Bouldin index (DB)

DB averages, over clusters, the worst similarity between a cluster and any other cluster [5, 96, 97]. Lower values indicate better clustering.

$$R_{i,j} = \frac{\frac{1}{n_i}\sum_{x \in C_i} d(x, \mu_i) + \frac{1}{n_j}\sum_{x \in C_j} d(x, \mu_j)}{d(\mu_i, \mu_j)}, \qquad \mathrm{DB} = \frac{1}{K}\sum_{i=1}^{K} \max_{i \neq j} R_{i,j}, \tag{A3}$$

where $\max_{i \neq j} R_{i,j}$ selects the most similar cluster to C_i. Because DB uses centroid-based compactness and separation, it may favour compact globular clusters.

Calinski-Harabasz index (CH)

Also known as Variance Ratio Criterion, it compares between-cluster dispersion with within-cluster dispersion [98, 99]. Higher values indicate better clustering, and the

index has no upper bound. It is defined as:

$$\mathrm{CH} = \frac{\mathrm{BSS}/(K-1)}{\mathrm{WSS}/(N-K)} = \frac{\mathrm{BSS}}{\mathrm{WSS}} \cdot \frac{N-K}{K-1}, \quad \mathrm{WSS} = \sum_{k=1}^{K} \sum_{x_i \in C_k} d(x_i, \mu_k)^2, \quad \mathrm{BSS} = \sum_{k=1}^{K} n_k d(\mu_k, \mu_{\mathrm{data}})^2, \tag{A4}$$

where WSS is the within-cluster sum of squares and BSS is the between-cluster sum of squares, as defined in (A1) and (A2), respectively, asuming $K > 1$, and $N > K$.

Silhouette index (S)

The Silhouette index compares, for each point, its mean distance to points in the same cluster with its mean distance to points in the nearest other cluster [98, 100]. Higher values indicate better clustering, with $S \in [-1, 1]$, having the definition:

$$a_i = \frac{1}{n_{c(i)} - 1} \sum_{\substack{x_j \in C_{c(i)} \\ j \neq i}} d(x_i, x_j), \quad b_i = \min_{l \neq c(i)} \left(\frac{1}{n_l} \sum_{x_j \in C_l} d(x_i, x_j) \right), \quad S = \frac{1}{N} \sum_{i=1}^{N} \frac{b_i - a_i}{\max(a_i, b_i)}. \tag{A5}$$

Here, $c(i)$ denotes the cluster containing x_i. Because S relies on distance-based compactness and separation, it may favour compact, well-separated clusters and can be sensitive to noise and outliers [12].

Centroid-based Silhouette index (cSIL)

The centroid-based Silhouette index is a variant of the Silhouette index in which point-to-cluster distances are computed using cluster centroids rather than average pairwise distances. Higher values indicate better clustering, with $\mathrm{cSIL} \in [-1, 1]$:

$$a_i^c = d(x_i, \mu_{c(i)}), \quad b_i^c = \min_{l \neq c(i)} d(x_i, \mu_l), \quad \mathrm{cSIL} = \frac{1}{N} \sum_{i=1}^{N} \frac{b_i^c - a_i^c}{\max(a_i^c, b_i^c)}, \tag{A6}$$

where $c(i)$ denotes the cluster containing x_i.

For reproducibility, the implementations of XB, GD43, GD53, PS, rCIP, and WB follow the CVI library.

Xie-Beni index (XB)

It evaluates the compactness and separation of a clustering solution by comparing total within-cluster compactness to the closest inter-cluster separation, originally developed for fuzzy clustering [101] Lower XB values indicate better clustering solutions (more compact clusters and larger separation). The hard-partition XB is computed as:

$$\mathrm{XB} = \frac{\sum_{k=1}^{K} \sum_{x \in C_k} d(x, \mu_k)^2}{N \min_{1 \leq i < j \leq K} d(\mu_i, \mu_j)^2}. \tag{A7}$$

Generalised Dunn's index 43 (GD43)

GD43 compares the minimum centroid separation with the largest within-cluster dispersion term [102, 103]. Higher values indicate better clustering, as seen in:

$$GD43 = \frac{\min_{1\le i<j\le K} d(\mu_i, \mu_j)}{\max_{1\le k\le K} \dfrac{2 \cdot \sum_{x\in C_k} d(x, \mu_k)^{1/2}}{n_k}}, \tag{A8}$$

where $\max(d(i,k)),\ k \neq j$ is the maximum distance between data point i and any other data point in a different cluster.

Generalised Dunn's index 53 (GD53)

It is similar in spirit to GD43, but replaces the centre-to-centre separation with a size-weighted average half-power distance between two clusters [102, 103]. Higher GD53 values also indicate better clustering. GD53 can be more robust than GD43 for non-spherical or differently sized clusters because it uses per-cluster half-power averages in the numerator rather than raw centre distances:

$$GD53 = \frac{\min_{1\le i<j\le K} \left(\dfrac{\sum_{x\in C_i} d(x, \mu_i)^{1/2} + \sum_{x\in C_j} d(x, \mu_j)^{1/2}}{n_i + n_j} \right)}{\max_{1\le k\le K} \dfrac{2 \cdot \sum_{x\in C_k} d(x, \mu_k)^{1/2}}{n_k}}. \tag{A9}$$

Partition Separation (PS)

It comprises a measure of separation between prototypes/centroids [104]. Larger values of PS indicate better clustering solutions (maximization).

PS evaluates prototype separation (not internal compactness) and outputs higher scores for well-spaced, balanced prototype partitions:

$$\mathrm{PS} = \sum_{k=1}^{K} \left(\frac{n_k}{\max_{k\in K} n_k} - \exp\left[-\frac{\min_{1\le i<j\le K} d(\mu_i, \mu_j)^2}{\frac{1}{K}\sum_{k=1}^{K} d(\mu_k, \mu_{\mathrm{data}})^2} \right] \right). \tag{A10}$$

Renyi's representative Cross Information Potential (rCIP)

It represents each cluster C_k by a Gaussian centred at the cluster centroid c_k with a (regularised) sample covariance Σ_k, then measures the pairwise overlap of those Gaussians [105]. rCIP aggregates the overlaps between all distinct cluster pairs: a small overlap means well-separated clusters. rCIP is non-negative (0 score means perfect separation) and lower values indicate better clustering:

$$\Sigma_k = \frac{1}{n_k - 1} \sum_{x\in C_k} (xx^\top - n_k \mu_k \mu_k^\top) + \varepsilon, \quad \mathrm{rCIP} = \sum_{1\le i<j\le K} (2\pi)^{-\frac{D}{2}} \cdot |\Sigma_i + \Sigma_j|^{-\frac{1}{2}} \cdot e^{-\frac{1}{2}(\mu_i - \mu_j)^T (\Sigma_i + \Sigma_j)^{-1} (\mu_i - \mu_j)}. \tag{A11}$$

WB-index (WB)

It measures how good a clustering result is by balancing within-cluster compactness and between-cluster separation [106]. Lower values generally indicate better clustering quality. WB considers average distances and not the shape or distribution of clusters, yet WB was found to be one of the most robust CVIs in a recent study [12]. It is defined by:

$$\mathrm{WB} = \frac{K \cdot \sum_{k=1}^{K} \sum_{x \in C_k} d(x, \mu_k)^2}{\sum_{k=1}^{K} n_k \cdot d(\mu_k, \mu_{\mathrm{data}})^2}. \tag{A12}$$

The following indices were implemented using the Permetrics library [107]: DBCV, $\mathcal{D}$, DH, SSE, RS, B, BH, and H.

Density-based clustering validation (DBCV)

DBCV is designed for density-based clustering results, such as those produced by DBSCAN [22, 108]. It compares within-cluster density sparseness with between-cluster density separation; higher values indicate better clustering, with DBCV $\in [-1, 1]$. For a point $x \in C_k$, the all-points-core-distance (APCD) is defined as:

$$\mathrm{APCD}(x) = \left(\frac{1}{n_k - 1} \sum_{i=1, x_i \neq x}^{n_k} \frac{1}{\mathrm{KNN}(x, i)^D} \right)^{-1/D}. \tag{A13}$$

The mutual reachability distance (MRD) between two objects x_i and x_j is:

$$\mathrm{MRD}(x_i, x_j) = \max\{\mathrm{APCD}(x_i), \mathrm{APCD}(x_j), d(x_i, x_j)\}. \tag{A14}$$

Minimum spanning trees (MSTs) are built for each cluster using MRDs. The density sparseness of a cluster (DSC) and the density separation between clusters (DSBC) are defined as:

$$\mathrm{DSC}(C_i) = \max\{\text{weight of internal edges in MST of } C_i\}; \tag{A15a}$$

$$\mathrm{DSBC}(C_i, C_j) = \min_{x_p \in C_i,\, x_q \in C_j} \mathrm{MRD}(x_p, x_q). \tag{A15b}$$

The validity of a cluster and the final DBCV index are calculated as:

$$\mathrm{DBVC}(C_i) = \frac{\min_{j \neq i}(\mathrm{DSBC}(C_i, C_j)) - \mathrm{DSC}(C_i)}{\max(\min_{j \neq i}(\mathrm{DSBC}(C_i, C_j)), \mathrm{DSC}(C_i))}, \quad \mathrm{DBCV} = \sum_{k=1}^{K} \frac{n_k}{N} \cdot \mathrm{DBVC}(C_k). \tag{A16}$$

DBCV captures density and shape information, but may be less reliable when clusters are connected by bridges or are not well separated [109].

Dunn index ($\mathcal{D}$)

The Dunn index compares the smallest inter-cluster separation with the largest intra-cluster diameter [102, 110]. Higher values indicate better clustering:

$$\Delta(C_k) = \max_{x_i, x_j \in C_k, i \neq j} d(x_i, x_j), \qquad \Delta(C_k, C_l) = \min_{x_i \in C_k, x_j \in C_l} d(x_i, x_j), \qquad \mathcal{D} = \frac{\min_{1 \leq k < l \leq K} \Delta(C_k, C_l)}{\max_{1 \leq k \leq K} \Delta(C_k)}. \tag{A17}$$

The Dunn index can be sensitive to scale and outliers [12].

Duda-Hart index (DH)

The Duda-Hart index compares average within-cluster pairwise distance with average between-cluster pairwise distance [111]. Lower values indicate better clustering:

$$D_{\text{intra}}(C_k) = \frac{2}{n_k(n_k - 1)} \sum_{x_i, x_j \in C_k, i < j} d(x_i, x_j), \qquad D_{\text{inter}}(C_k, C_l) = \frac{1}{n_k n_l} \sum_{x_i \in C_k, x_j \in C_l} d(x_i, x_j), \tag{A18a}$$

$$\text{DH} = \frac{\frac{1}{K} \sum_{k=1}^{K} D_{\text{intra}}(C_k)}{\frac{2}{K(K-1)} \sum_{k=1}^{K} \sum_{l=k+1}^{K} D_{\text{inter}}(C_k, C_l)}. \tag{A18b}$$

Sum of Squared Error index (SSE)

SSE measures within-cluster compactness as the total squared distance from points to their assigned cluster centroid [112]. Lower values indicate more compact clusters:

$$\text{SSE} = \sum_{k=1}^{K} \sum_{x \in C_k} d(x, \mu_k)^2. \tag{A19}$$

The R-squared index (RS)

The R-squared index measures the proportion of total dispersion explained by the clustering solution. Higher values indicate better clustering:

$$\text{RS} = 1 - \frac{\text{SSE}}{\text{SSE} + \text{BSS}}. \tag{A20}$$

As a variance-based measure, RS is primarily suited to compact, centroid-based cluster structures.

Beale index (B)

The Beale index is an F-ratio-type criterion comparing between-cluster and within-cluster sums of squares [113, 114]. Higher values indicate better clustering:

$$B = \frac{\text{BSS}/(K-1)}{\text{SSE}/(N-K)}. \tag{A21}$$

Ball-Hall index (BH)

The Ball-Hall index measures average within-cluster compactness through squared point-to-centroid distances [113, 115]. Lower values indicate more compact clusters:

$$BH = \frac{1}{N} \sum_{k=1}^{K} \sum_{x \in C_k} d(x, \mu_k)^2. \tag{A22}$$

Hartigan index (H)

The Hartigan index compares within-cluster compactness with the distance from points to the nearest external cluster centroid [116]. Lower values indicate better clustering:

$$H = \frac{\sum_{k=1}^{K} \sum_{x \in C_k} d(x, \mu_k)^2}{\sum_{k=1}^{K} \sum_{x \in C_k} d(x, \mu_l)^2}, \tag{A23}$$

where μ_l denotes the centroid of the closest cluster distinct from C_k.

The following indices were implemented using the PyCVI library [117]: SF, SD, SDbw, XB*, COP, C, I, Sym, CDbw, VIASCKDE, CS.

Score Function (SF)

The Score Function combines a between-cluster distance term (BCD) and a within-cluster distance term (WCD), mapping their difference to $(0, 1)$ [118]. Higher values indicate better clustering:

$$\mathrm{BCD} = \frac{1}{N} \sum_{k=1}^{K} n_k \cdot d(\mu_k, \mu_{\mathrm{data}}), \quad \mathrm{WCD} = \sum_{k=1}^{K} \left(\frac{1}{n_k} \cdot \sum_{x \in C_k} d(x, \mu_k) \right), \quad \mathrm{SF} = 1 - \frac{1}{e^{e^{(\mathrm{BCD} - \mathrm{WCD})}}}. \tag{A24}$$

SD validity index (SD)

The SD index combines cluster scattering with centroid-based separation [119]. Lower values indicate better clustering:

$$\mathrm{Scat}_{\mathrm{SD}} = \frac{1}{K} \sum_{k=1}^{K} \frac{\left(\sum_{q=1}^{D} \sigma_{kq}^2 \right)^{1/2}}{\left(\sum_{q=1}^{D} \sigma_q^2 \right)^{1/2}}, \quad \sigma_{kq}^2 = \frac{1}{n_k} \sum_{x_i \in C_k} (x_{iq} - \mu_{kq})^2, \quad \sigma_q^2 = \frac{1}{N} \sum_{i=1}^{N} (x_{iq} - \mu_{\mathrm{data},q})^2. \tag{A25a}$$

$$\rho_{kl} = d(\mu_k, \mu_l), \quad \rho_{\min} = \min_{1 \le k < l \le K} \rho_{kl}, \quad \rho_{\max} = \max_{1 \le k < l \le K} \rho_{kl}, \quad \mathrm{Dis}_{\mathrm{SD}} = \frac{\rho_{\max}}{\rho_{\min}} \sum_{k=1}^{K} \left(\sum_{l=1,\, l \neq k}^{K} \rho_{kl} \right)^{-1}, \tag{A25b}$$

$$\mathrm{SD} = \alpha_{\mathrm{SD}} \cdot \mathrm{Scat}_{\mathrm{SD}} + \mathrm{Dis}_{\mathrm{SD}}. \tag{A25c}$$

Here, σ_{kq} is the standard deviation of cluster C_k along feature q, σ_q is the standard deviation of the full dataset along feature q, and α_{SD} is a scaling coefficient, commonly set from the maximum number of clusters considered.

S_ Dbw validity index (SDbw)

The SDbw index combines cluster scattering with density between clusters [120]. Lower values indicate better clustering:

$$r_{\mathrm{SDbw}} = \frac{1}{K}\left(\sum_{k=1}^{K}\|\sigma_k\|\right)^{1/2}, \quad \phi_r(t) = \begin{cases} 1, & \text{if } t \le r_{\mathrm{SDbw}}, \\ 0, & \text{otherwise}, \end{cases} \tag{A26a}$$

$$\mathrm{dens}(u, C_k) = \sum_{x \in C_k} \phi_r(d(u,x)), \qquad m_{kl} = \frac{\mu_k + \mu_l}{2}, \tag{A26b}$$

$$\mathrm{Dens}_{\mathrm{bw}} = \frac{2}{K(K-1)} \sum_{k=1}^{K-1} \sum_{l=k+1}^{K} \frac{\mathrm{dens}(m_{kl}, C_k) + \mathrm{dens}(m_{kl}, C_l)}{\max\{\mathrm{dens}(\mu_k, C_k), \mathrm{dens}(\mu_l, C_l)\}}; \tag{A26c}$$

$$\mathrm{SDbw} = \mathrm{Scat}_{\mathrm{SDbw}} + \mathrm{Dens}_{\mathrm{bw}}. \tag{A26d}$$

Here, σ_k denotes the feature-wise standard deviation vector of cluster C_k, and m_{kl} is the midpoint between centroids μ_k and μ_l. Because SDbw is centroid- and density-based, it can be sensitive to overlap, outliers, and arbitrary-shaped clusters [12, 22].

Xie-Beni index (XB*)*

XB* is a variant of the Xie-Beni index that uses the worst per-cluster average dispersion relative to the minimum centroid separation [49]. Lower values indicate better clustering:

$$\mathrm{XB}^* = \frac{\max\limits_{1 \le k \le K} \frac{1}{n_k} \sum\limits_{x \in C_k} d(x, \mu_k)^2}{\min\limits_{1 \le k < l \le K} d(\mu_k, \mu_l)^2}. \tag{A27}$$

COP index (COP)

The COP index compares average point-to-centroid compactness with a farthest-neighbor separation term [121]. Lower values indicate better clustering, with $\mathrm{COP} \in [0, \infty)$:

$$\mathrm{COP} = \frac{1}{K} \sum_{k=1}^{K} \frac{\frac{1}{n_k} \sum_{x \in C_k} d(x, \mu_k)}{\min\limits_{x \notin C_k} \max\limits_{y \in C_k} d(x, y)}. \tag{A28}$$

Because COP uses centroid-based compactness, it may be sensitive to non-globular cluster shapes.

C index (C)

The C index compares the observed sum of within-cluster pairwise distances with the minimum and maximum possible sums over the same number of pairs [122]. It is

based on the assumption that the clusters are spherical, equally sized, and have the same density. Lower values indicate better clustering:

$$S_{\text{within}} = \sum_{k=1}^{K} \sum_{x_i, x_j \in C_k,\ i<j} d(x_i, x_j), \quad C = \frac{S_{\text{within}} - S_{\min}}{S_{\max} - S_{\min}}. \tag{A29}$$

Here, $S_{\min}$ and $S_{\max}$ are the sums of the m smallest and m largest pairwise distances in the dataset, respectively, where $m = \sum_{k=1}^{K} \binom{n_k}{2}$ is the number of within-cluster pairs.

I index (I)

The I index combines global data dispersion, within-cluster compactness, and maximum centroid separation [53]. Higher values indicate better clustering:

$$\mathrm{I} = \frac{\sum_{i=1}^{N} d(x_i, \mu_{\text{data}})}{\mathrm{SSE}} \cdot \max_{1 \le k < l \le K} d(\mu_k, \mu_l). \tag{A30}$$

Because it uses centroid-based separation and SSE, it may favour compact, convex clusters.

Sym index (Sym)

The Sym index extends the I index by replacing the compactness term with a point-symmetry-based distance, favouring clusters whose points are symmetrically arranged around their representatives. Higher values indicate better clustering. For a point $x_i \in C_k$, let $x_i^{(k)\prime} = 2\mu_k - x_i$ be the reflection of x_i with respect to the centroid μ_k. The point-symmetry distance is defined as:

$$d_{\text{sym}}(x_i, \mu_k) = d_E(x_i, \mu_k) \cdot \frac{1}{q} \sum_{x_j \in \mathcal{N}_q(x_i^{(k)\prime})} d_E(x_i^{(k)\prime}, x_j), \tag{A31}$$

where $\mathcal{N}_q(x_i^{(k)\prime})$ denotes the set of the q nearest neighbors of the reflected point. The Sym index is then:

$$\mathrm{Sym} = \frac{\sum_{i=1}^{N} d_{\text{sym}}(x_i, \mu_{\text{data}})}{\sum_{k=1}^{K} \sum_{x_i \in C_k} d_{\text{sym}}(x_i, \mu_k)} \cdot \max_{1 \le k < l \le K} d_E(\mu_k, \mu_l). \tag{A32}$$

CDbw index (CDbw)

CDbw combines compactness with a density-penalised separation term and was designed to better evaluate arbitrarily shaped clusters [39]. Higher values indicate better clustering:

$$f_\sigma(x, u) = \begin{cases} 1, & d(x,u) < \sigma, \\ 0, & \text{otherwise}, \end{cases}, \quad \mathrm{Dens}(C_k, C_l) = \frac{d(\mu_k, \mu_l)}{2\sigma} \cdot \mathrm{card}(\mu_{kl}), \tag{A33a}$$

$$\text{InterDens} = \frac{1}{K}\sum_{k=1}^{K}\max_{l\neq k}\text{Dens}(C_k, C_l), \quad \text{Separation} = \frac{1}{K}\sum_{k=1}^{K}\frac{\min_{l\neq k} d(\mu_k, \mu_l)}{1+\text{InterDens}}, \tag{A33b}$$

$$\text{Cohesion} = \frac{\text{Compactness}}{1+\text{IntraChange}}, \quad \text{CDbw} = \text{Cohesion}\cdot\text{Separation}\cdot\text{Compactness}. \tag{A33c}$$

Because CDbw relies on density estimates and cluster representatives, its score can depend on representative selection and neighborhood parameters, especially for clusters with different sizes, densities, or bridge-like connections [22, 109].

VIASCKDE index (VIASCKDE)

VIASCKDE evaluates clustering through pointwise compactness and separation rather than centroid-based distances, and was designed for arbitrary-shaped clusters [43]. Higher values indicate better clustering. For a point $x_i \in C_{c(i)}$, define:

$$a_i = \min_{\substack{x_j \in C_{c(i)} \\ j\neq i}} d(x_i, x_j), \quad b_i = \min_{l\neq c(i)} \min_{x_j\in C_l} d(x_i, x_j), \tag{A34}$$

where $c(i)$ denotes the cluster containing x_i. The pointwise compactness-separation score is:

$$\text{CoSeD}(x_i) = w_{\text{KDE}}(x_i)\cdot\frac{b_i - a_i}{\max(a_i, b_i)}, \tag{A35}$$

where $w_{\text{KDE}}(x_i)$ is the KDE-based weight of point x_i. The final index is:

$$\text{VIASCKDE} = \tfrac{1}{N}\textstyle\sum_{i=1}^{N}\text{CoSeD}(x_i). \tag{A36}$$

CS index (CS)

The CS index compares within-cluster scatter with centroid-based inter-cluster separation, similarly to DB and the Dunn index ($\mathcal{D}$) [38].

$$\text{CS} = \frac{\sum_{k=1}^{K}\left(\frac{1}{n_k}\sum_{x_i\in C_k}\max_{x_j\in C_k} d(x_i, x_j)\right)}{\sum_{k=1}^{K}\min_{1\leq l\leq K, l\neq k} d(\mu_k, \mu_l)}. \tag{A37}$$

Appendix B Execution time analysis

In this subsection, we present a dual execution time analysis of all CVIs for a simple dataset, containing 3 clusters with different standard deviations, with varying number of samples (Table B1) and number of dimensions (Table B2). It can be seen that the three best performing CVIs that have also been developed for irregular cluster shapes, specifically DBCV, VIASCKDE, and CDbw, have a significantly higher execution time than the proposed methods. DBCV and VIASCKDE have a steep rise in execution time as the number of samples increase. The proposed methods suffer from the same phenomenon; however, the execution times are significantly lower.

Specifically, CDbw has the lowest execution time when considering only the number of samples in a fixed $D = 2$ setup; however, when considering varying number of dimensions, its issues emerge as for $D = 50$ it is unable to be computed while for $D = 10$ the execution time increased exponentially. DBCV remains somewhat consistent regardless of the number of dimensions, yet it has a high execution time of approximately 40 s, while VIASCKDE has an exponential increase as dimensions become higher. For the proposed methods, an increase in time can be seen as the number of dimensions increases up until $D = 50$, after which the execution time remains consistent (for $D = 100$ and $D = 500$).

Table B1: Comparison of execution times in seconds (mean ± standard deviation) of CVIs for varying number of samples (with fixed 2 dimensions).

	n=500	n=1000	n=5000	n=10000	n=15000	n=20000
cSIL	0.001±0.000	0.002±0.000	0.008±0.000	0.017±0.000	0.026±0.000	0.031±0.001
GD43	0.000±0.001	0.001±0.001	0.002±0.000	0.004±0.000	0.007±0.001	0.010±0.002
GD53	0.001±0.001	0.000±0.001	0.002±0.001	0.004±0.001	0.009±0.002	0.010±0.002
PS	0.000±0.001	0.001±0.001	0.002±0.001	0.004±0.001	0.007±0.002	0.008±0.001
rCIP	0.001±0.001	0.001±0.001	0.002±0.001	0.006±0.002	0.007±0.001	0.009±0.002
WB	0.000±0.000	0.001±0.001	0.002±0.001	0.004±0.000	0.007±0.000	0.008±0.000
XB	0.000±0.000	0.000±0.001	0.002±0.001	0.004±0.000	0.008±0.002	0.008±0.001
SSE	0.000±0.000	0.000±0.000	0.000±0.001	0.001±0.000	0.002±0.000	0.002±0.000
RS	0.000±0.000	0.000±0.000	0.001±0.000	0.001±0.000	0.002±0.001	0.003±0.001
DH	0.002±0.001	0.008±0.000	0.208±0.023	1.157±0.083	1.911±0.061	3.278±0.428
B	0.000±0.000	0.000±0.000	0.001±0.000	0.002±0.000	0.003±0.000	0.003±0.000
BH	0.000±0.000	0.000±0.000	0.001±0.001	0.001±0.001	0.001±0.000	0.001±0.000
$\mathcal{D}$	0.000±0.001	0.001±0.001	0.015±0.001	0.059±0.004	0.129±0.004	0.208±0.002
H	0.000±0.000	0.000±0.001	0.001±0.001	0.002±0.001	0.004±0.001	0.004±0.001
DBCV	4.580±0.082	5.864±0.285	8.326±0.459	14.104±0.520	26.849±1.872	50.311±3.856
I	0.000±0.000	0.000±0.000	0.001±0.000	0.002±0.002	0.004±0.003	0.006±0.008
C	0.006±0.000	0.031±0.002	0.920±0.094	4.145±0.361	8.499±0.410	16.517±0.803
CDbw	0.007±0.002	0.008±0.001	0.013±0.001	0.023±0.001	0.041±0.027	0.063±0.029
VIASCKDE	0.080±0.010	0.268±0.027	5.963±0.270	23.994±1.714	54.283±1.365	99.118±2.332
COP	0.005±0.001	0.008±0.000	0.040±0.001	0.075±0.002	0.131±0.010	0.246±0.043
Sym	0.001±0.000	0.002±0.001	0.010±0.001	0.018±0.000	0.044±0.021	0.041±0.003
CS	0.000±0.001	0.001±0.000	0.030±0.001	0.104±0.004	0.284±0.019	0.441±0.029
SF	0.000±0.000	0.001±0.000	0.002±0.000	0.005±0.001	0.011±0.008	0.012±0.004
SD	0.014±0.000	0.052±0.004	1.006±0.038	3.804±0.022	8.578±0.301	14.767±0.244
SDbw	0.001±0.000	0.002±0.001	0.005±0.000	0.010±0.000	0.014±0.001	0.019±0.001
XB*	0.000±0.001	0.001±0.000	0.002±0.000	0.005±0.001	0.007±0.001	0.009±0.001
S	0.004±0.000	0.019±0.001	0.328±0.004	1.285±0.027	2.222±0.012	3.829±0.013
DB	0.001±0.000	0.002±0.000	0.001±0.000	0.002±0.000	0.004±0.001	0.006±0.007
CH	0.000±0.000	0.001±0.000	0.001±0.000	0.001±0.000	0.002±0.000	0.002±0.001
AD-S	0.016±0.001	0.040±0.002	0.868±0.023	2.328±0.024	4.883±0.266	8.876±0.822
AD-DB	0.017±0.001	0.038±0.002	0.857±0.022	2.482±0.121	5.409±0.403	8.657±0.411
AD-CH	0.017±0.003	0.045±0.014	0.900±0.050	2.572±0.014	4.822±0.148	8.932±0.452
AD-idea	0.028±0.003	0.074±0.007	1.073±0.031	3.385±0.190	6.998±0.115	12.331±0.358

Table B2: Comparison of execution times in seconds (mean ± standard deviation) of CVIs for varying number of dimensions (with fixed 1000 samples).

	d=2	d=5	d=10	d=50	d=100	d=500
cSIL	0.004±0.000	0.003±0.001	0.005±0.001	0.003±0.000	0.005±0.001	0.024±0.001
GD43	0.001±0.000	0.001±0.000	0.002±0.000	0.001±0.000	0.001±0.000	0.008±0.001
GD53	0.001±0.000	0.001±0.000	0.001±0.000	0.001±0.001	0.001±0.000	0.007±0.001
PS	0.002±0.000	0.001±0.000	0.001±0.000	0.001±0.000	0.001±0.000	0.003±0.000
rCIP	0.002±0.001	0.002±0.000	0.003±0.001	0.011±0.001	0.044±0.024	NaN±NaN
WB	0.001±0.001	0.001±0.000	0.001±0.001	0.001±0.000	0.001±0.000	0.008±0.001
XB	0.001±0.001	0.001±0.000	0.001±0.000	0.001±0.001	0.001±0.001	0.008±0.001
SSE	0.000±0.000	0.000±0.000	0.001±0.001	0.001±0.000	0.000±0.000	0.006±0.001
RS	0.000±0.000	0.000±0.000	0.000±0.000	0.000±0.000	0.001±0.000	0.006±0.001
DH	0.014±0.002	0.017±0.003	0.022±0.001	0.039±0.001	0.061±0.002	0.289±0.007
B	0.001±0.001	0.001±0.000	0.001±0.000	0.000±0.000	0.001±0.000	0.007±0.001
BH	0.000±0.000	0.000±0.000	0.000±0.000	0.000±0.000	0.001±0.000	0.005±0.001
$\mathcal{D}$	0.002±0.001	0.002±0.000	0.003±0.001	0.005±0.001	0.010±0.000	0.053±0.003
H	0.001±0.001	0.000±0.000	0.001±0.000	0.001±0.001	0.001±0.000	0.005±0.001
DBCV	35.616±1.897	37.947±3.115	40.591±1.398	39.638±2.062	37.895±8.462	34.312±0.446
I	0.000±0.000	0.001±0.001	0.000±0.001	0.001±0.000	0.001±0.001	0.010±0.000
CDbw	0.744±0.091	0.998±0.227	1209±167	`NaN`	`NaN`	`NaN`
C	0.039±0.001	0.066±0.010	0.059±0.003	0.055±0.005	0.065±0.002	0.198±0.018
VIASCKDE	0.306±0.017	0.532±0.068	0.711±0.017	0.863±0.093	1.305±0.053	6.013±0.311
COP	0.010±0.001	0.029±0.004	0.017±0.004	0.010±0.001	0.012±0.001	0.016±0.001
Sym	0.002±0.001	0.018±0.003	0.063±0.006	0.235±0.013	0.498±0.023	2.246±0.069
CS	0.002±0.000	0.004±0.001	0.006±0.001	0.011±0.000	0.019±0.001	0.093±0.004
SF	0.001±0.001	0.001±0.000	0.002±0.000	0.001±0.000	0.001±0.000	0.005±0.000
SD	0.029±0.003	0.046±0.005	0.077±0.006	0.087±0.005	0.157±0.007	0.799±0.004
SDbw	0.001±0.000	0.003±0.001	0.005±0.001	0.010±0.001	0.015±0.001	0.094±0.004
XB*	0.001±0.001	0.001±0.000	0.001±0.000	0.001±0.000	0.001±0.000	0.006±0.000
S	0.017±0.002	0.020±0.003	0.025±0.002	0.020±0.001	0.020±0.001	0.027±0.001
DB	0.001±0.000	0.001±0.000	0.002±0.001	0.003±0.001	0.004±0.001	0.007±0.002
CH	0.000±0.000	0.000±0.001	0.001±0.000	0.001±0.000	0.001±0.000	0.006±0.001
AD-S	0.050±0.007	0.084±0.046	0.362±0.016	0.233±0.006	0.241±0.005	0.249±0.006
AD-DB	0.050±0.006	0.063±0.004	0.349±0.065	0.233±0.005	0.241±0.004	0.251±0.006
AD-CH	0.064±0.008	0.077±0.023	0.283±0.011	0.232±0.004	0.238±0.003	0.245±0.003
AD-idea	0.106±0.007	0.204±0.100	0.351±0.006	0.445±0.008	0.453±0.011	0.485±0.022

Appendix C Additional application: AD-K-Means

The Arboris Distance can also be used as a replacement for Euclidean distance in clustering algorithms. Another use case is in the K-Means [46] clustering algorithm, as it is known to have issues with irregularly shaped clusters and imbalanced clusters [123]. The analysis compares K-Means, DBSCAN, HDBSCAN, MeanShift, AgglomerativeClustering, SpectralClustering and the ED- and AD- extensions of K-Means from the perspective of ARI (Table C3), AMI (TableC4) and execution time (Table C5) on six datasets: unbalance [66], aggregation [56], compound [57], jain [62], pathbased and spiral [63].

Parameter values were found with a grid search for DBSCAN, HDBSCAN, MeanShift, AgglomerativeClustering, and SpectralClustering due to their high number of

complex parameters. AD-K-Means has a single parameter $n_{\text{neighbors}}$ (excluding the k of K-Means), which, as shown above, does not significantly affect the results, making it much easier to use.

The scores obtained by ARI (Table C3) and AMI (Table C4) for these clustering algorithms indicate that the AD-extension of K-Means outperforms K-Means on all datasets and SpectralClustering and MeanShift on almost all datasets. However, DBSCAN, HDBSCAN and AgglomerativeClustering obtain better scores in most cases.

Table C3: Comparison of clustering algorithms by AMI.

Algorithm	aggregation	compound	jain	pathbased	spiral	unbalance
K-Means	0.838	0.725	0.526	0.543	-0.005	1
DBSCAN	0.983	0.908	1	0.887	1	0.999
HDBSCAN	0.898	0.839	0.934	0.725	1	1
MeanShift	0.889	0.796	0.433	0.578	0.236	1
AgglomerativeClustering	0.928	0.813	0.928	0.563	1	0.974
SpectralClustering	0.774	0.862	0.667	0.713	1	0.796
ED-K-Means	0.818	0.746	1	0.524	0.404	0.923
AD-K-Means	0.701	0.835	1	0.490	1	1

Table C4: Comparison of clustering algorithms by ARI.

Algorithm	aggregation	compound	jain	pathbased	spiral	unbalance
K-Means	0.735	0.583	0.577	0.461	-0.006	1
DBSCAN	0.987	0.949	1	0.92	1	1
HDBSCAN	0.827	0.883	0.976	0.682	1	1
MeanShift	0.868	0.77	0.367	0.574	0.113	1
AgglomerativeClustering	0.929	0.796	0.976	0.485	1	0.995
SpectralClustering	0.539	0.807	0.733	0.657	1	0.61
ED-K-Means	0.652	0.567	1	0.449	0.353	0.886
AD-K-Means	0.795	0.826	1	0.430	1	1

Appendix D Balancing external CVIs

External CVIs may also be affected by the number of predicted clusters [34] and by class imbalance [35, 36]. To reduce the effect of imbalance, we define balanced variants of external CVIs that give equal importance to each ground-truth class.

The approach decomposes the global comparison into independent one-vs-rest evaluations. For each ground-truth class, the predicted cluster that best matches that class is selected, the chosen external CVI is computed on the corresponding binary labels, and one score is obtained per class, as shown in Algorithm 4. The final score

Table C5: Comparison of clustering algorithms by execution time in seconds.

Algorithm	aggregation	compound	jain	pathbased	spiral	unbalance
K-Means	0.047	0.346	0.051	0.019	0.245	0.034
DBSCAN	0.008	0.003	0.004	0.003	0.003	0.343
HDBSCAN	0.01	0.006	0.006	0.007	0.008	0.125
MeanShift	0.178	0.147	0.127	0.086	0.098	0.25
AgglomerativeClustering	0.019	0.007	0.002	0.001	0.001	0.612
SpectralClustering	0.126	0.062	0.091	0.027	0.027	1.013
ED-K-Means	96.392	23.45	3.839	2.055	5.959	3820.879
AD-K-Means	0.058	0.022	0.046	0.052	0.021	3.438s

is then obtained by aggregating the per-class scores, for example by macro, weighted, or harmonic averaging.

Algorithm 4 Balancing function for external CVIs

function BALANCEDCVI($CVI, true_labels, pred_labels$)
  $classes \leftarrow$ unique values in $true_labels$
  if size($classes$) $= 1$ **then**
    return 1.0
  end if
  $class_scores \leftarrow$ empty list
  for each c **in** $classes$ **do**
    $mask \leftarrow (true_labels == c)$
    $true_binary \leftarrow mask$ as $\{0, 1\}$
    $pred_subset \leftarrow pred_labels[mask]$
    $most_common_cluster \leftarrow$ argmax_count($pred_subset$)
    $pred_binary \leftarrow (pred_labels == most_common_cluster)$ as $\{0, 1\}$
    $score_c \leftarrow CVI(true_binary, pred_binary)$
    append $score_c$ **to** $class_scores$
  end for
  return mean($class_scores$)
end function

Several studies use external CVIs to assess or compare internal CVIs through correlations or related analyses [20, 22]. However, when the external CVI is itself sensitive to class imbalance, these comparisons may give an incomplete view of performance. In highly imbalanced datasets, errors affecting small classes may contribute little to the global score, even when they are important for the interpretation of the clustering. To illustrate this effect, we generated a sequence of imbalanced datasets by extracting cluster statistics from the unbalanced dataset [66] and progressively decreasing the imbalance.

We compare the ground-truth labels with a labelset in which the minority clusters are randomly assigned. These minority clusters correspond to the five smaller clusters

on the right side of the dataset. Figure D1 shows that ARI assigns an almost perfect score when the errors occur mainly in these minority clusters, because the majority classes dominate the pairwise comparisons. In contrast, Balanced ARI (BARI) assigns a lower score because each ground-truth class contributes equally to the final evaluation. When the minority clusters are upsampled to match the majority clusters (Figure D1, bottom row), ARI and BARI produce nearly identical scores, confirming that the discrepancy is driven by class imbalance.

In this work, we use macro averaging, i.e., the arithmetic mean of per-class scores, so that each class contributes equally regardless of its sample size. This choice is intentionally simple and directly targets imbalance in the ground-truth labels.

This balancing strategy has trade-offs. Equal class weighting makes the metric more sensitive to noise in very small classes, where a small number of mislabelled points can strongly affect the class-level score. In addition, selecting only the most-overlapping predicted cluster may underrepresent cases where a true class is split across several predicted clusters, i.e., over-clustering. Thus, balanced external CVIs are most useful when evaluation should reflect per-class performance, such as in fairness-sensitive analyses, rare-class detection, or small-group evaluation.

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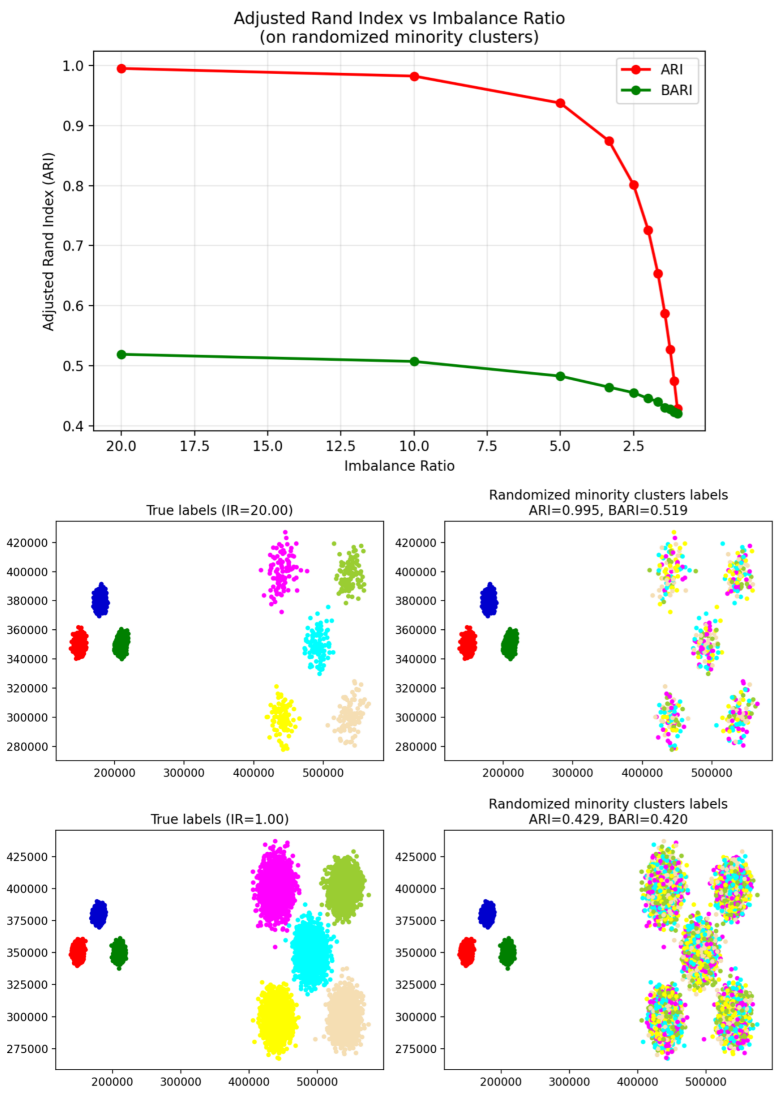


Fig. D1: Comparison of external CVIs: ARI and Balanced ARI (BARI)

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